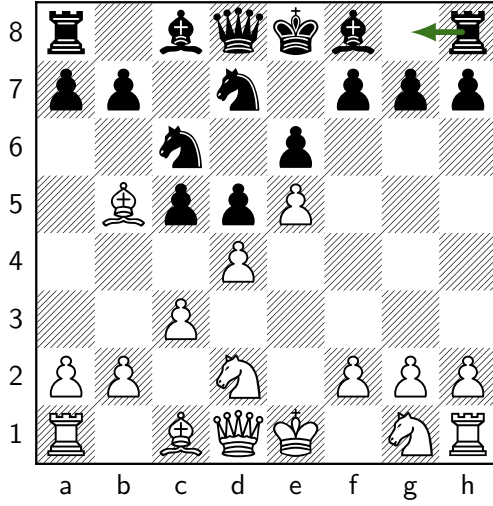


Melanie-ANALOGY — French, Tarrasch

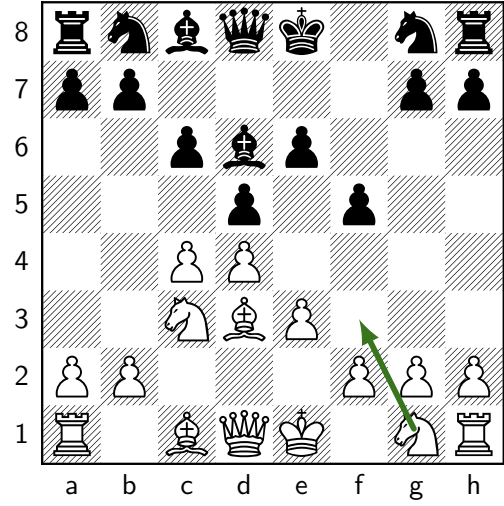
Melanie (Deep_Vision)

Abstract

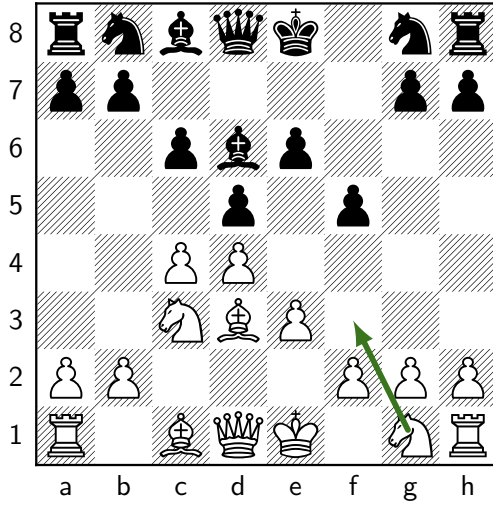
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



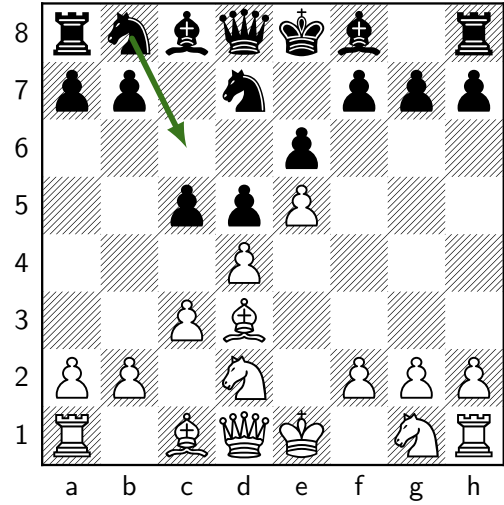
(a)



(b)

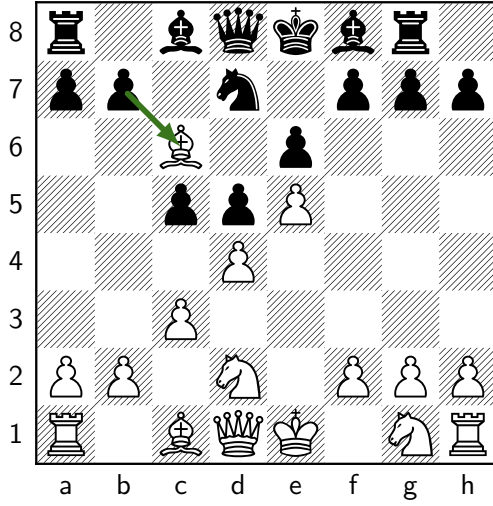


(c)

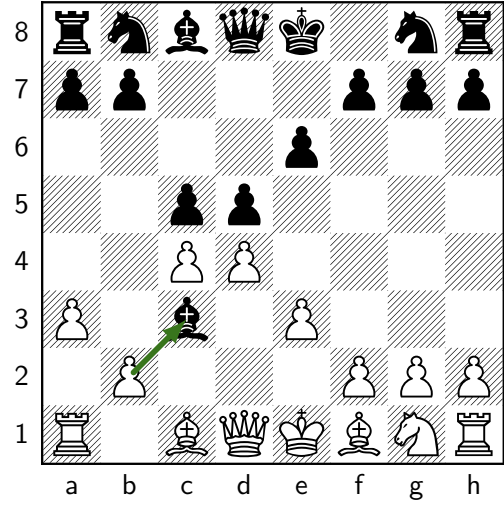


(d)

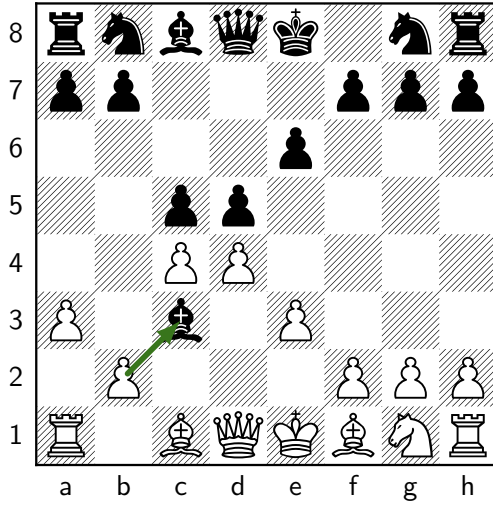
Figure 1: **Move 7, Black: Rg8**. (a) Our position; Melanie plays Rg8 (h8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1193` (ply 10) [structural distance $d=72$, QAP obj=57]: the master played Nf3; under μ our *Rh1* plays the role of that game’s *Ng1* (a **cross-type** correspondence); cross-type roles: our *Bf8* as their *Ra1*, our *ra1* as their *Bc1*; transferred move h8g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1194` (ply 10) [structural distance $d=72$, QAP obj=57]: the master played Nf3; under μ our *Rh1* plays the role of that game’s *Ng1* (a **cross-type** correspondence); cross-type roles: our *Bf8* as their *Ra1*, our *ra1* as their *Bc1*; transferred move h8g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3114` (ply 11) [structural distance $d=42$, QAP obj=32]: the master played Nc6; under μ our *Rh1* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Bf8* as their *Ra8*, our *Pe6* as their *Bf8*; transferred move h8g8 (**endorses this move**). Synthesis: 20 retrieved analogs transferred to Rg8 (summed QAP objective 694, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



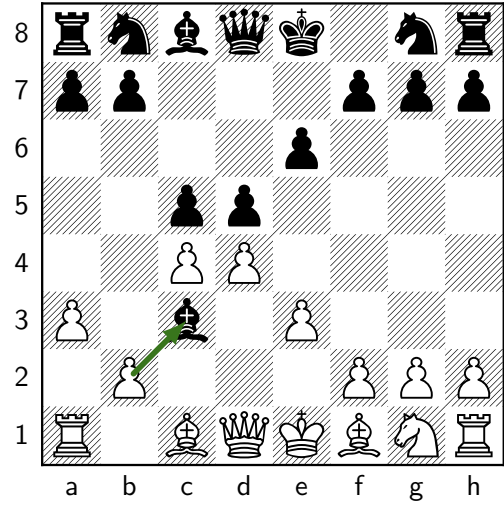
(a)



(b)

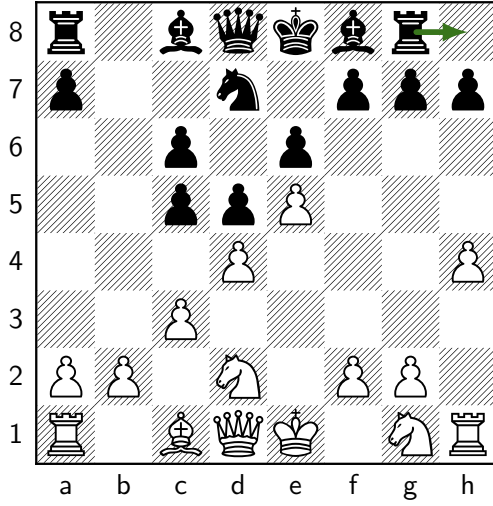


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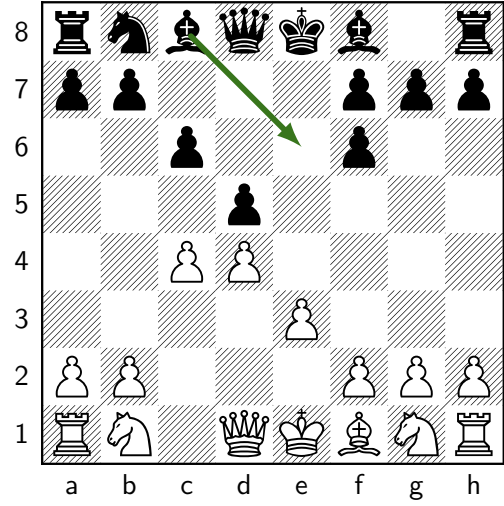


(d)

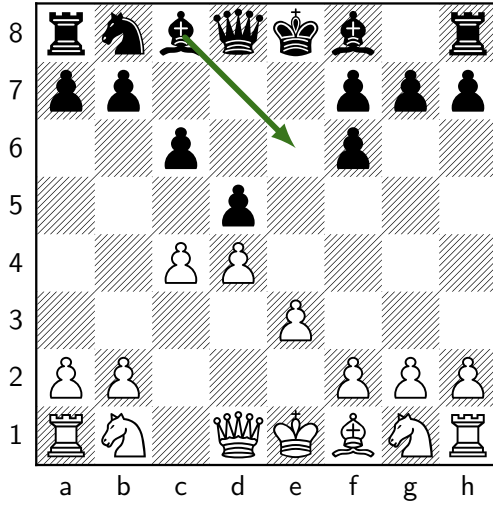
Figure 2: **Move 8, Black: bxc6**. (a) Our position; Melanie plays **bxc6** (**b7c6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1241` (ply 10) [structural distance $d=63$, QAP obj=19]: the master played **bxc3**; under μ our *Pb2* plays the role of that game’s *Pb2*; cross-type roles: our *Ph7* as their *Ra1*, our *pd4* as their *Bf1*; transferred move **b7c6** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1242` (ply 10) [structural distance $d=63$, QAP obj=19]: the master played **bxc3**; under μ our *Pb2* plays the role of that game’s *Pb2*; cross-type roles: our *Ph7* as their *Ra1*, our *pd4* as their *Bf1*; transferred move **b7c6** (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1858` (ply 10) [structural distance $d=63$, QAP obj=19]: the master played **bxc3**; under μ our *Pb2* plays the role of that game’s *Pb2*; cross-type roles: our *Ph7* as their *Ra1*, our *pd4* as their *Bf1*; transferred move **b7c6** (**endorses this move**). Synthesis: 6 retrieved analogs transferred to **bxc6** (summed QAP objective 112, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



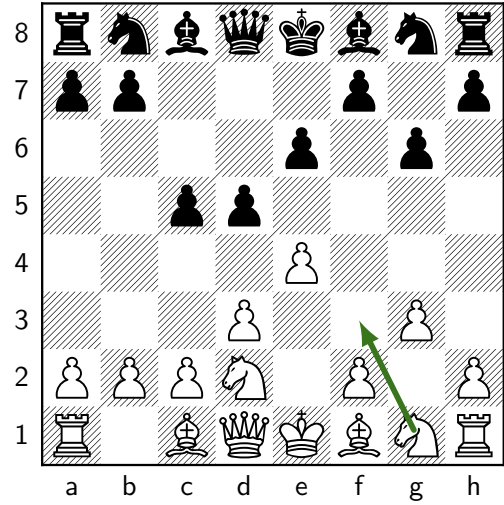
(a)



(b)

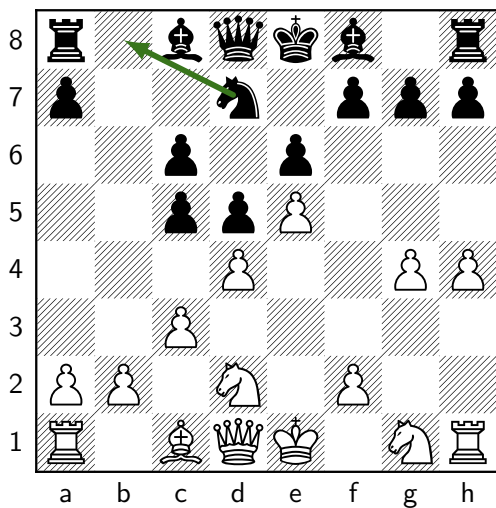


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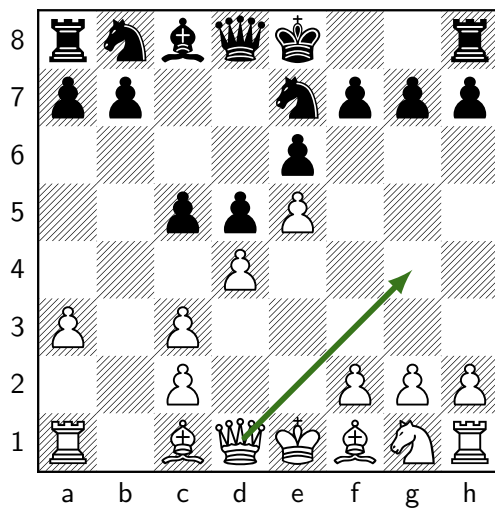


(d)

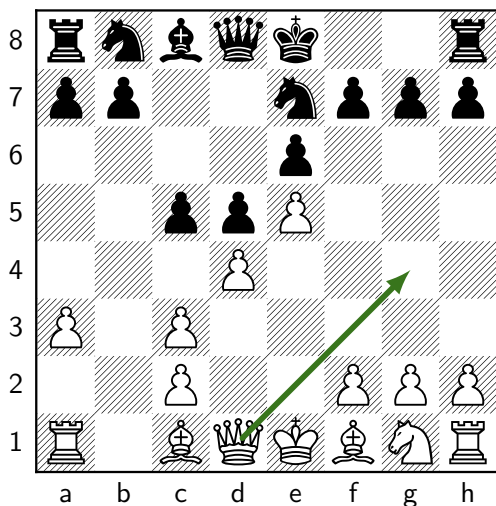
Figure 3: **Move 9, Black: Rh8.** (a) Our position; Melanie plays Rh8 (g8h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#458` (ply 9) [structural distance $d=71$, QAP obj=44]: the master played Be6; under μ our *Rg1* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *pg2* as their *Ra8*, our *Bc8* as their *Nb8*; transferred move g8h8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#459` (ply 9) [structural distance $d=71$, QAP obj=44]: the master played Be6; under μ our *Rg1* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *pg2* as their *Ra8*, our *Bc8* as their *Nb8*; transferred move g8h8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1394` (ply 8) [structural distance $d=74$, QAP obj=43]: the master played Ngf3; under μ our *Rg1* plays the role of that game’s *Ng1* (a **cross-type** correspondence); cross-type roles: our *Pc6* as their *Bf1*, our *ph4* as their *Rh1*; transferred move g8h8 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Rh8 (summed QAP objective 275, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



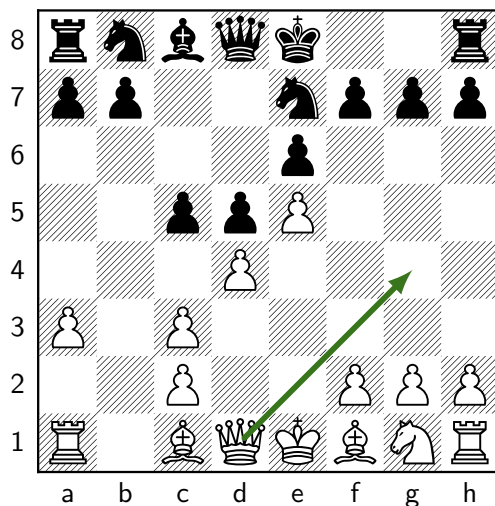
(a)



(b)

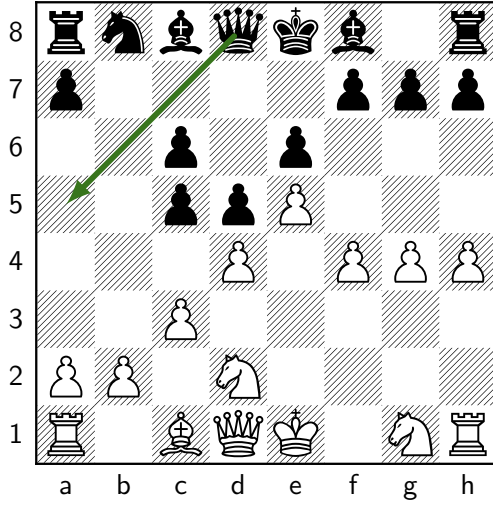


(c)

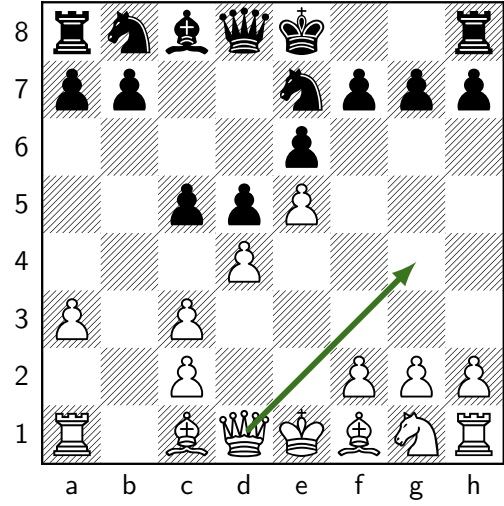


(d)

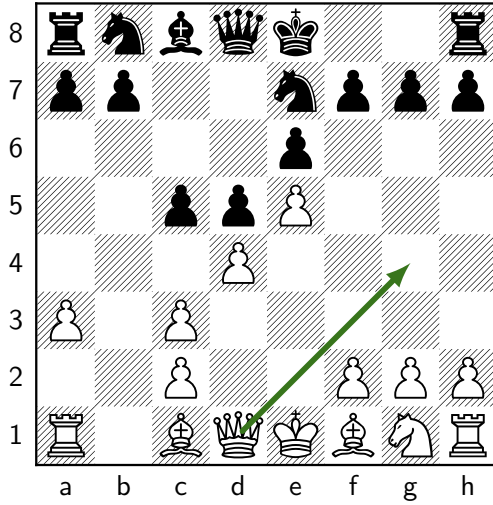
Figure 4: **Move 10, Black: Nb8.** (a) Our position; Melanie plays Nb8 (d7b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1244` (ply 12) [structural distance $d=74$, QAP obj=45]: the master played Qg4; under μ our Nd2 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our pe5 as their Ra1, our Qd8 as their Bc1; transferred move d7b8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1245` (ply 12) [structural distance $d=74$, QAP obj=45]: the master played Qg4; under μ our Nd2 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our pe5 as their Ra1, our Qd8 as their Bc1; transferred move d7b8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1320` (ply 12) [structural distance $d=74$, QAP obj=45]: the master played Qg4; under μ our Nd2 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our pe5 as their Ra1, our Qd8 as their Bc1; transferred move d7b8 (**endorses this move**). Synthesis: 29 retrieved analogs transferred to Nb8 (summed QAP objective 1307, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



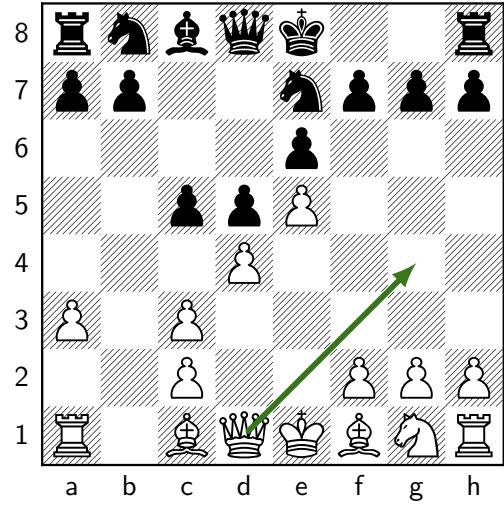
(a)



(b)

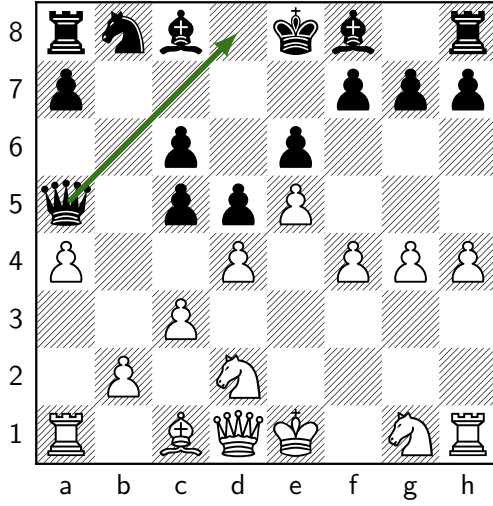


(c)

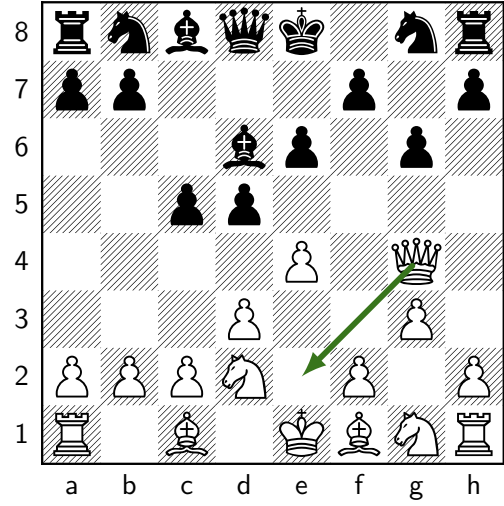


(d)

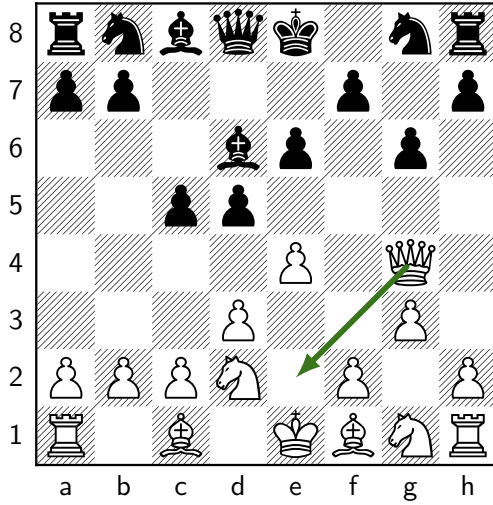
Figure 5: **Move 11, Black: Qa5.** (a) Our position; Melanie plays Qa5 (d8a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1244` (ply 12) [structural distance $d=59$, QAP obj=44]: the master played Qg4; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our rh1 as their Bc1, our pa2 as their Bf1; transferred move d8a5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1245` (ply 12) [structural distance $d=59$, QAP obj=44]: the master played Qg4; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our rh1 as their Bc1, our pa2 as their Bf1; transferred move d8a5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1320` (ply 12) [structural distance $d=59$, QAP obj=44]: the master played Qg4; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our rh1 as their Bc1, our pa2 as their Bf1; transferred move d8a5 (**endorses this move**). Synthesis: 37 retrieved analogs transferred to Qa5 (summed QAP objective 1610, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



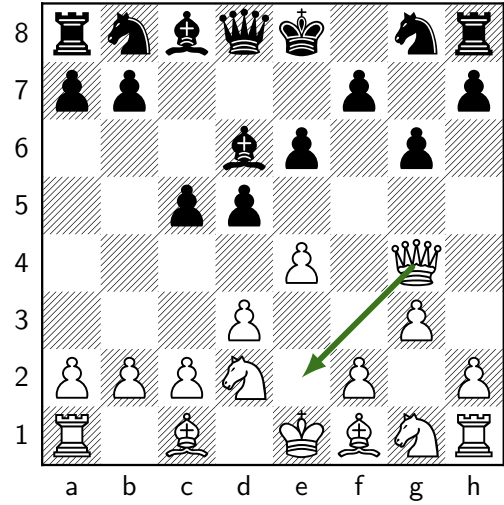
(a)



(b)

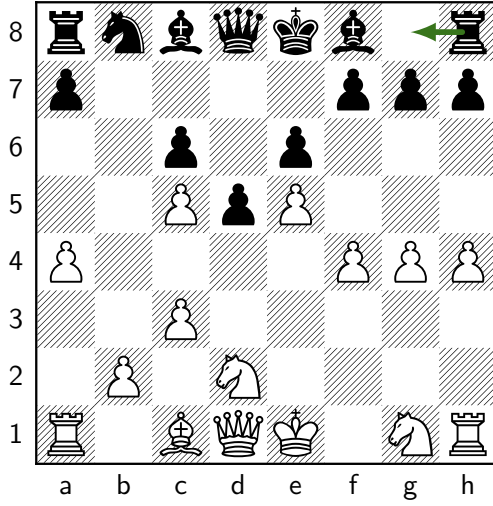


(c)

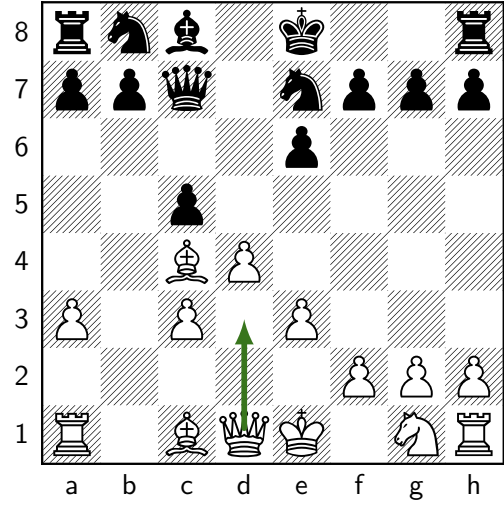


(d)

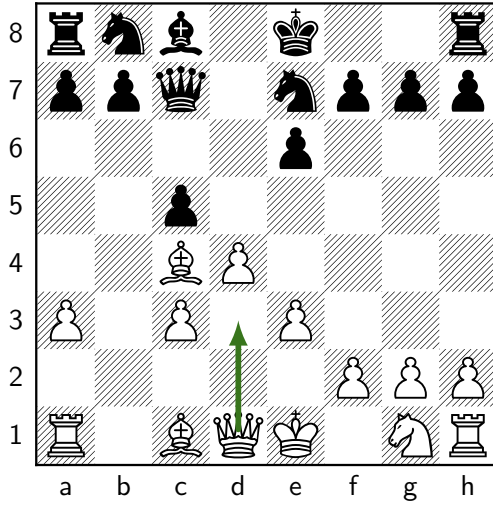
Figure 6: **Move 12, Black: Qd8.** (a) Our position; Melanie plays Qd8 (a5d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#19` (ply 10) [structural distance $d=74$, QAP obj=136]: the master played Qe2; under μ our Qa4 plays the role of that game’s Qg4; cross-type roles: our Rh8 as their Bc1, our Nb8 as their Bf1; transferred move a5d8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#20` (ply 10) [structural distance $d=74$, QAP obj=136]: the master played Qe2; under μ our Qa4 plays the role of that game’s Qg4; cross-type roles: our Rh8 as their Bc1, our Nb8 as their Bf1; transferred move a5d8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3828` (ply 10) [structural distance $d=74$, QAP obj=136]: the master played Qe2; under μ our Qa4 plays the role of that game’s Qg4; cross-type roles: our Rh8 as their Bc1, our Nb8 as their Bf1; transferred move a5d8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qd8 (summed QAP objective 545, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



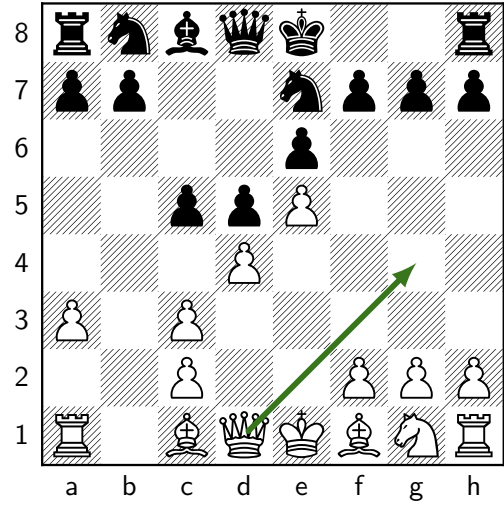
(a)



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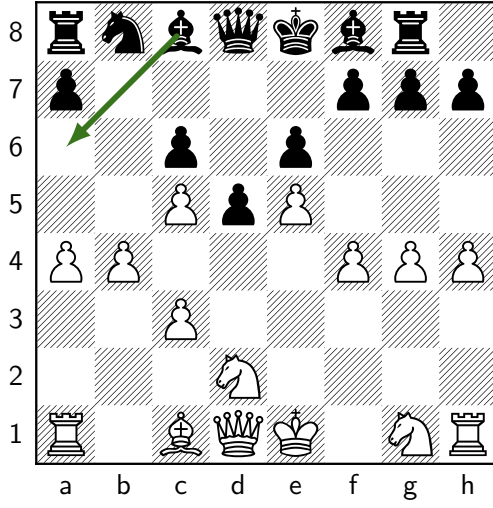


(c)

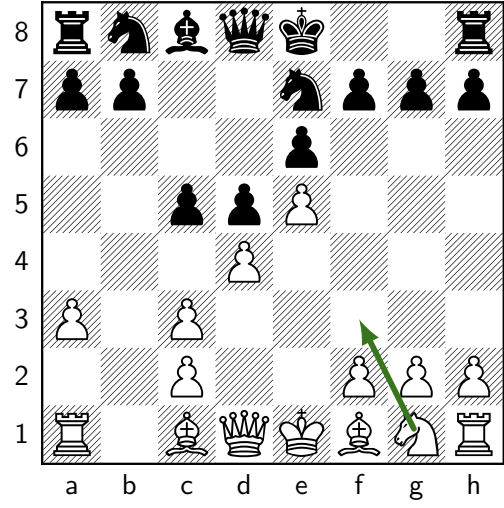


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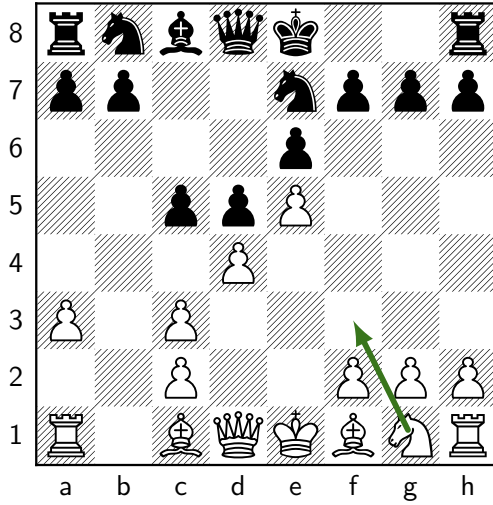
Figure 7: **Move 13, Black: Rg8.** (a) Our position; Melanie plays Rg8 (h8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1241` (ply 16) [structural distance $d=68$, QAP obj=15]: the master played Qd3; under μ our *Rh1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *pe5* as their *Ra1*, our *Pd5* as their *Bc1*; transferred move h8g8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1242` (ply 16) [structural distance $d=68$, QAP obj=15]: the master played Qd3; under μ our *Rh1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *pe5* as their *Ra1*, our *Pd5* as their *Bc1*; transferred move h8g8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1244` (ply 12) [structural distance $d=55$, QAP obj=12]: the master played Qg4; under μ our *Rh1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *pe5* as their *Ra1*, our *pf4* as their *Bf1*; transferred move h8g8 (**endorses this move**). Synthesis: 43 retrieved analogs transferred to Rg8 (summed QAP objective 508, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



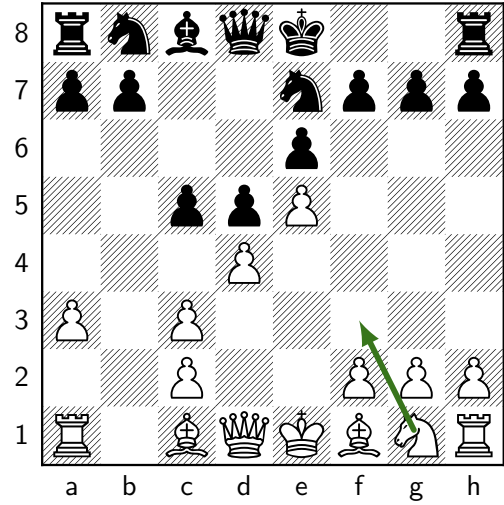
(a)



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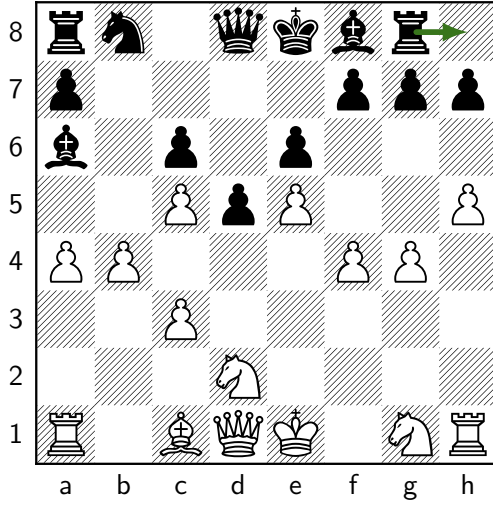


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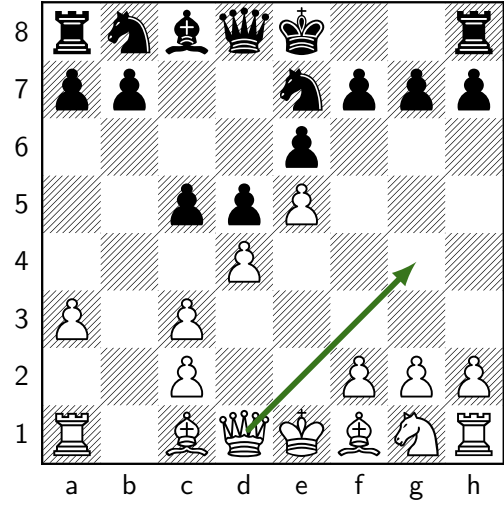


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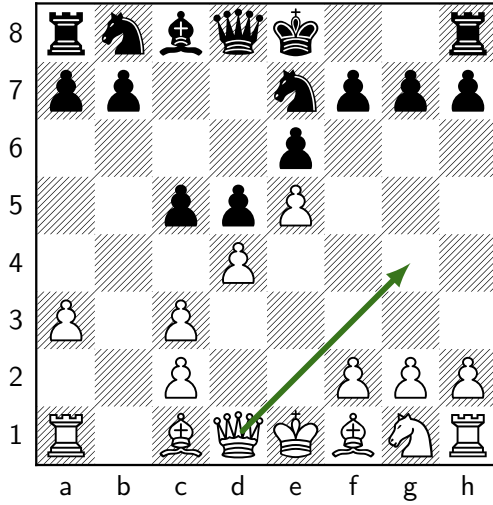
Figure 8: **Move 14, Black: Ba6.** (a) Our position; Melanie plays Ba6 (c8a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2700` (ply 12) [structural distance $d=69$, QAP obj=12]: the master played Nf3; under μ our Bc1 plays the role of that game’s Ng1 (a **cross-type** correspondence); cross-type roles: our Pc6 as their Ra1, our pe5 as their Bc1; transferred move c8a6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2701` (ply 12) [structural distance $d=69$, QAP obj=12]: the master played Nf3; under μ our Bc1 plays the role of that game’s Ng1 (a **cross-type** correspondence); cross-type roles: our Pc6 as their Ra1, our pe5 as their Bc1; transferred move c8a6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2726` (ply 12) [structural distance $d=69$, QAP obj=12]: the master played Nf3; under μ our Bc1 plays the role of that game’s Ng1 (a **cross-type** correspondence); cross-type roles: our Pc6 as their Ra1, our pe5 as their Bc1; transferred move c8a6 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Ba6 (summed QAP objective 140, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



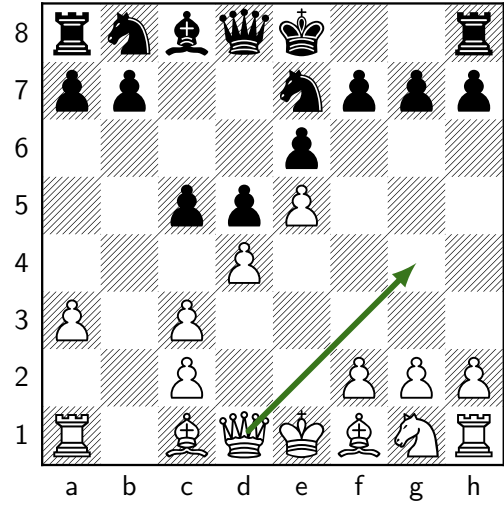
(a)



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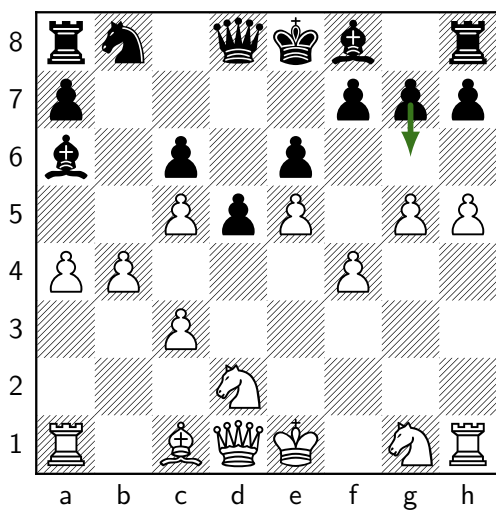


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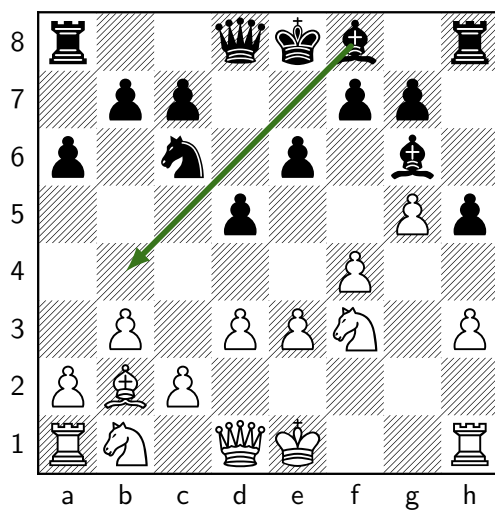


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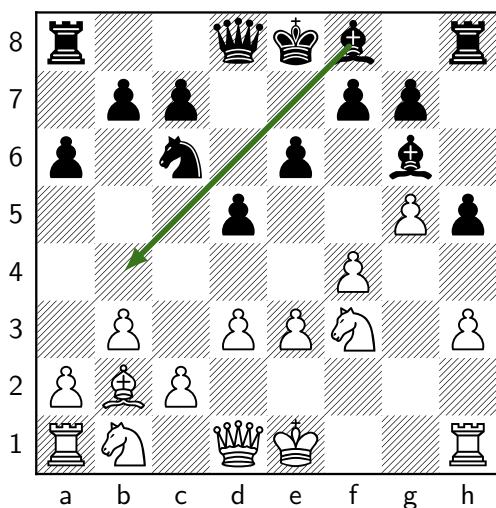
Figure 9: **Move 15, Black: Rh8.** (a) Our position; Melanie plays Rh8 (g8h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1244` (ply 12) [structural distance $d=101$, QAP obj=13]: the master played Qg4; under μ our *Rg1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *pe5* as their *Ra1*, our *pf4* as their *Bf1*; transferred move g8h8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1245` (ply 12) [structural distance $d=101$, QAP obj=13]: the master played Qg4; under μ our *Rg1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *pe5* as their *Ra1*, our *pf4* as their *Bf1*; transferred move g8h8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1320` (ply 12) [structural distance $d=101$, QAP obj=13]: the master played Qg4; under μ our *Rg1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *pe5* as their *Ra1*, our *pf4* as their *Bf1*; transferred move g8h8 (**endorses this move**). Synthesis: 38 retrieved analogs transferred to Rh8 (summed QAP objective 488, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



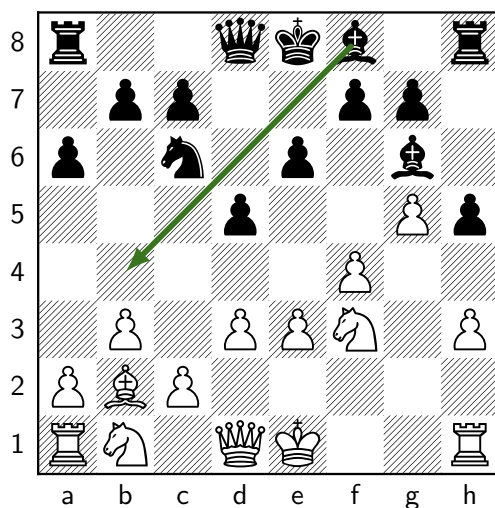
(a)



(b)

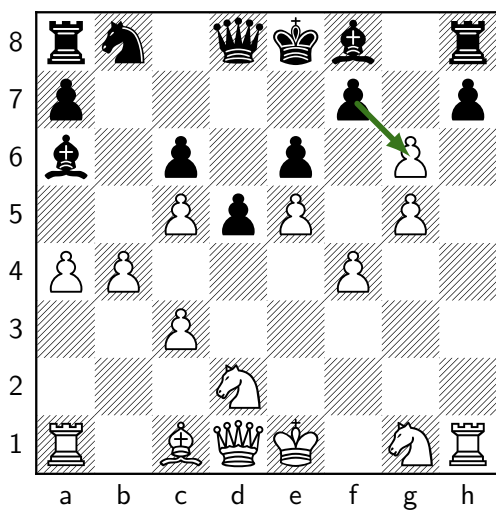


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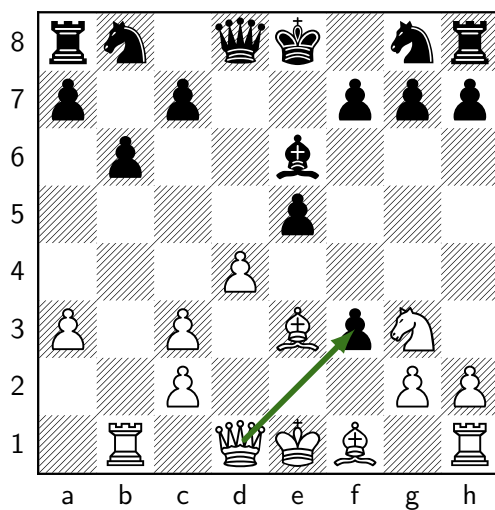


(d)

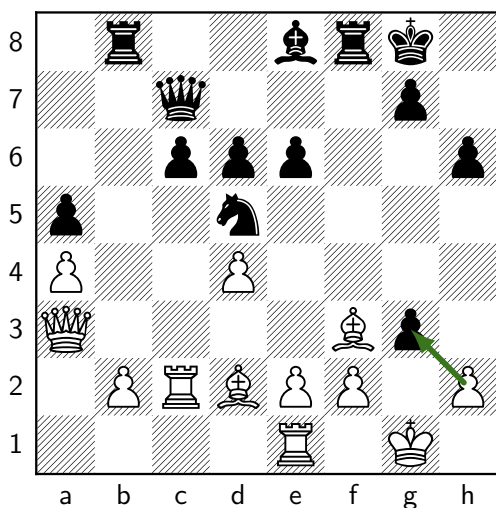
Figure 10: **Move 16, Black: g6** ($g7g6$). (a) Our position; Melanie plays $g6$ ($g7g6$). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1946` (ply 21) [structural distance $d=118$, QAP obj=51]: the master played $Bb4+$; under μ our $Pg2$ plays the role of that game’s $Bf1$ (a **cross-type** correspondence); cross-type roles: our $pa4$ as their $Rh8$, our $Nb8$ as their $Pc7$; transferred move $g7g6$ (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2753` (ply 21) [structural distance $d=118$, QAP obj=51]: the master played $Bb4+$; under μ our $Pg2$ plays the role of that game’s $Bf1$ (a **cross-type** correspondence); cross-type roles: our $pa4$ as their $Rh8$, our $Nb8$ as their $Pc7$; transferred move $g7g6$ (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2754` (ply 21) [structural distance $d=118$, QAP obj=51]: the master played $Bb4+$; under μ our $Pg2$ plays the role of that game’s $Bf1$ (a **cross-type** correspondence); cross-type roles: our $pa4$ as their $Rh8$, our $Nb8$ as their $Pc7$; transferred move $g7g6$ (**endorses this move**). Synthesis: 5 retrieved analogs transferred to $g6$ (summed QAP objective 200, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



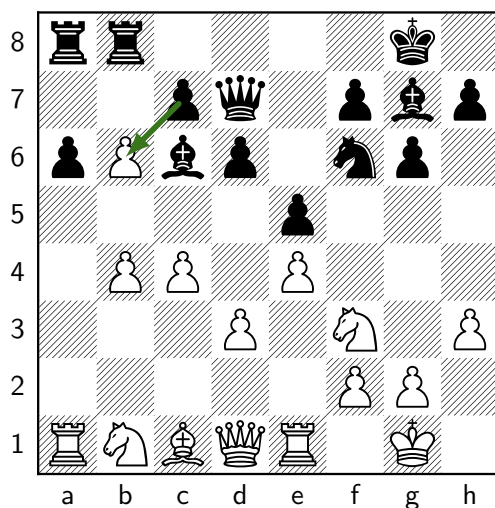
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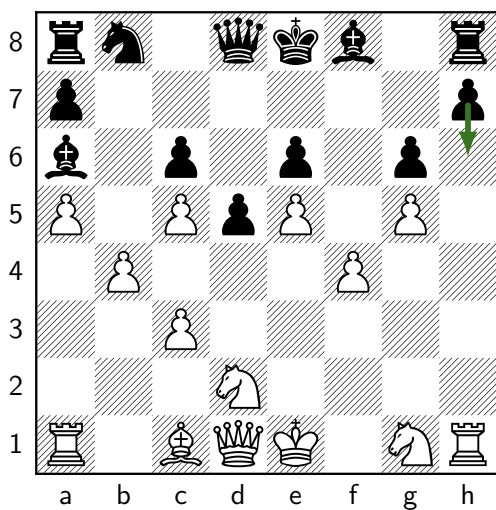


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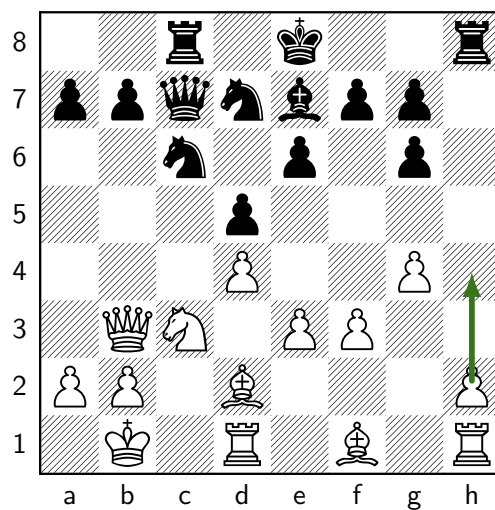


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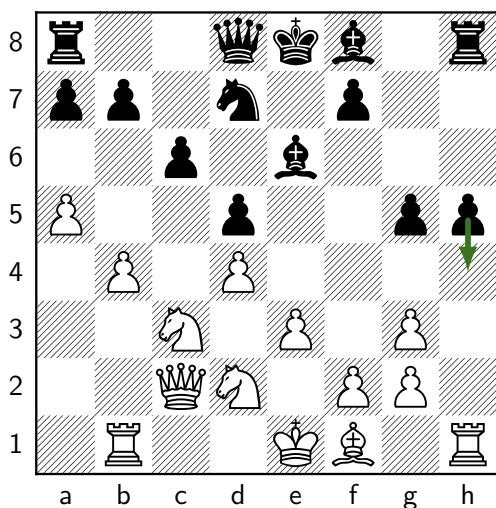
Figure 11: **Move 17, Black: fxg6.** (a) Our position; Melanie plays fxg6 (f7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1183` (ply 20) [structural distance $d=126$, QAP obj=113]: the master played Qxf3; under μ our Pf2 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Ba6 as their Rh1, our Nb8 as their Ph2; transferred move f7g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2165` (ply 46) [structural distance $d=141$, QAP obj=95]: the master played hxg3; under μ our Pf2 plays the role of that game’s Ph2; cross-type roles: our Qd8 as their Re1, our ng1 as their Pb2; transferred move f7g6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3419` (ply 27) [structural distance $d=133$, QAP obj=79]: the master played cxb6; under μ our Pf2 plays the role of that game’s Pc2; cross-type roles: our Ph7 as their Rb8, our Bf8 as their Nf6; transferred move f7g6 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to fxg6 (summed QAP objective 421, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



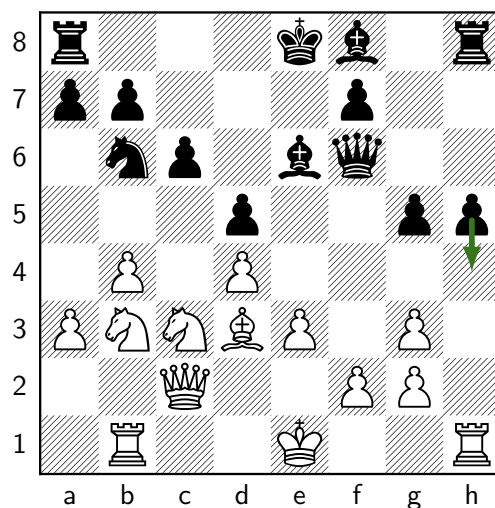
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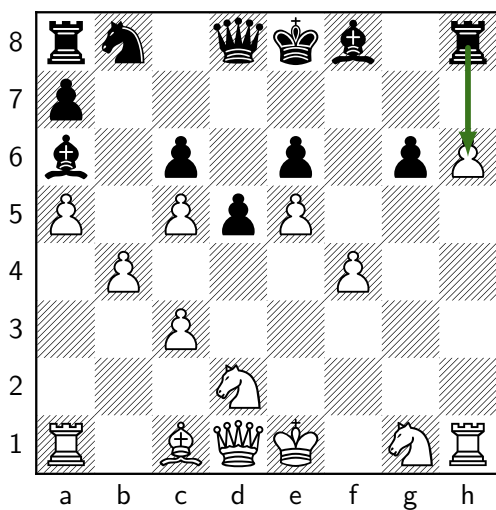


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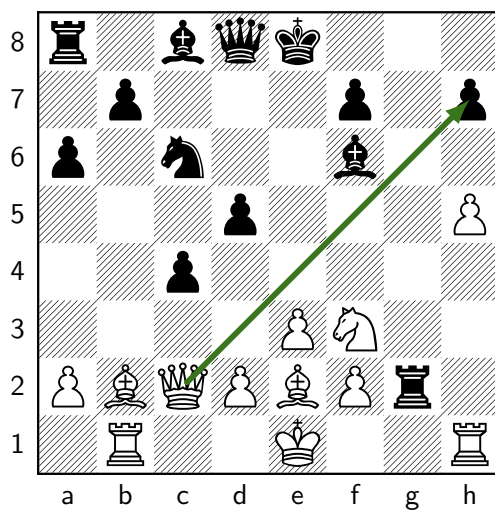


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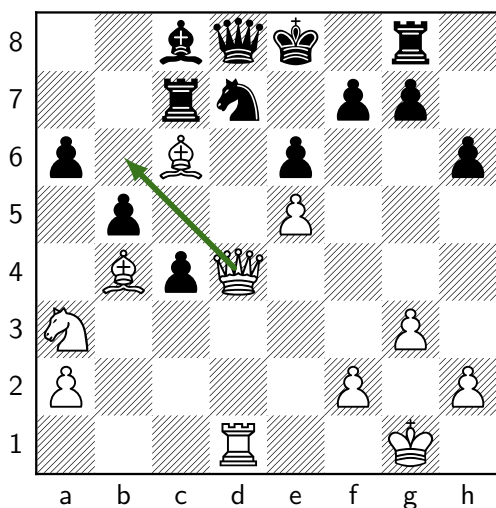
Figure 12: **Move 18, Black: h6 (h7h6).** (a) Our position; Melanie plays h6 (h7h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3459` (ply 28) [structural distance $d=129$, QAP obj=84]: the master played h4; under μ our $Ph2$ plays the role of that game’s $Ph2$; cross-type roles: our $pc3$ as their $Rd1$, our $Pe6$ as their $Bf1$; transferred move h7h6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3091` (ply 33) [structural distance $d=131$, QAP obj=63]: the master played h4; under μ our $Ph2$ plays the role of that game’s $Ph4$; cross-type roles: our $Ba6$ as their $Pb7$, our $pg5$ as their $Nd7$; transferred move h7h6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#518` (ply 33) [structural distance $d=123$, QAP obj=54]: the master played h4; under μ our $Ph2$ plays the role of that game’s $Ph4$; cross-type roles: our $Pe6$ as their $Ra8$, our $Qd8$ as their $Bf8$; transferred move h7h6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to h6 (summed QAP objective 235, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



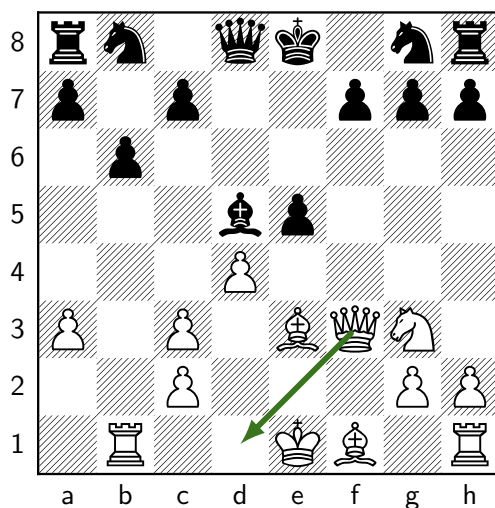
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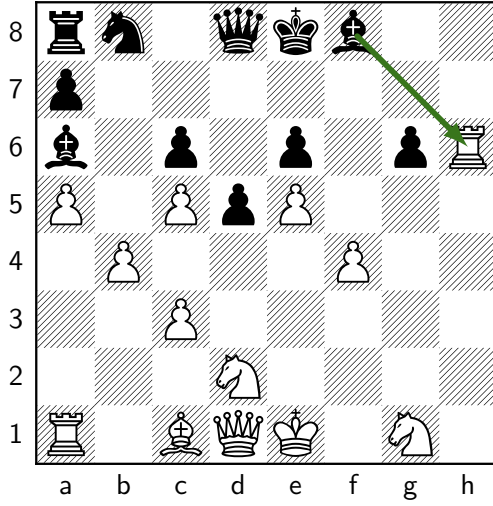


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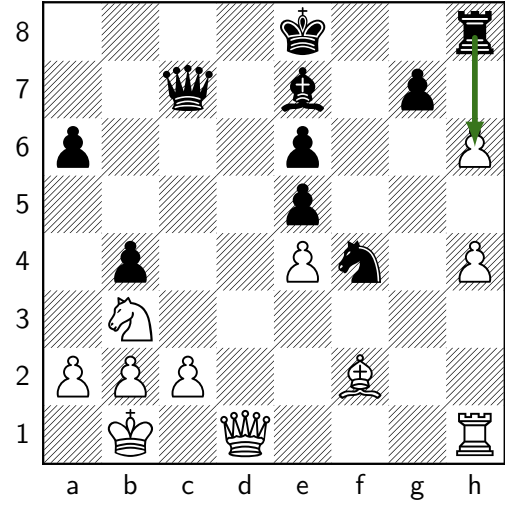


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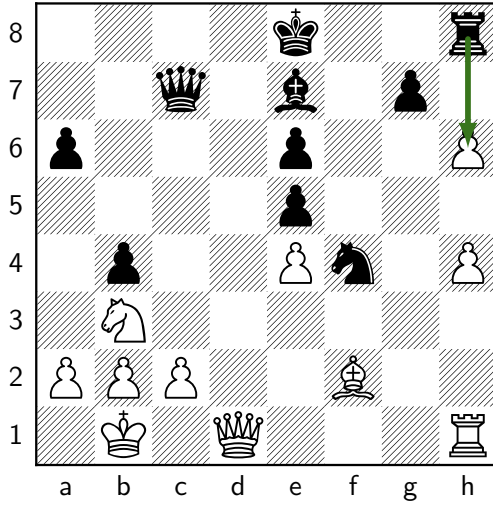
Figure 13: **Move 19, Black: Rxf6.** (a) Our position; Melanie plays Rxf6 (h8h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#856` (ply 30) [structural distance $d=131$, QAP obj=54]: the master played Qxh7; under μ our *Rh1* plays the role of that game’s *Qc2* (a **cross-type** correspondence); cross-type roles: our *Pa7* as their *Rb1*, our *Qd8* as their *Bb2*; transferred move h8h6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1408` (ply 48) [structural distance $d=119$, QAP obj=14]: the master played Qb6; under μ our *Rh1* plays the role of that game’s *Qd4* (a **cross-type** correspondence); cross-type roles: our *Qd8* as their *Rd1*, our *Ra8* as their *Ph2*; transferred move h8h6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1183` (ply 22) [structural distance $d=103$, QAP obj=12]: the master played Qd1; under μ our *Rh1* plays the role of that game’s *Qf3* (a **cross-type** correspondence); cross-type roles: our *Qd8* as their *Rb1*, our *nd2* as their *Rh1*; transferred move h8h6 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Rxf6 (summed QAP objective 95, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



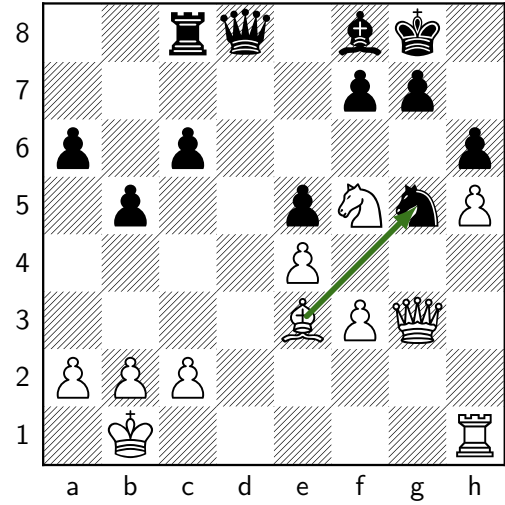
(a)



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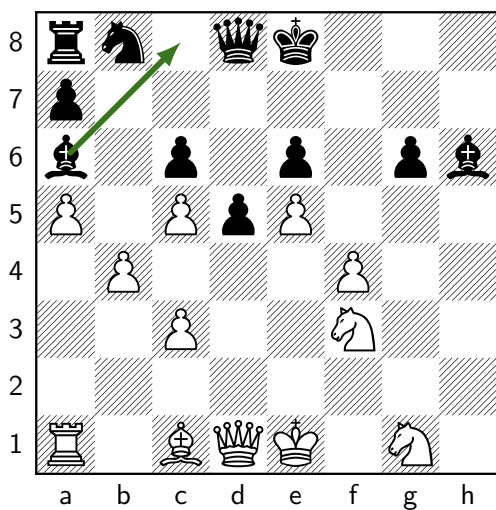


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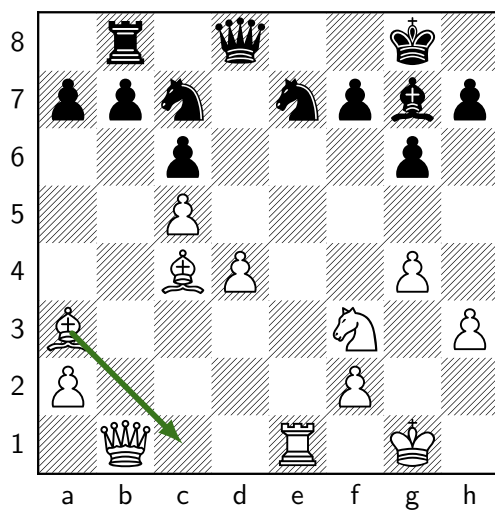


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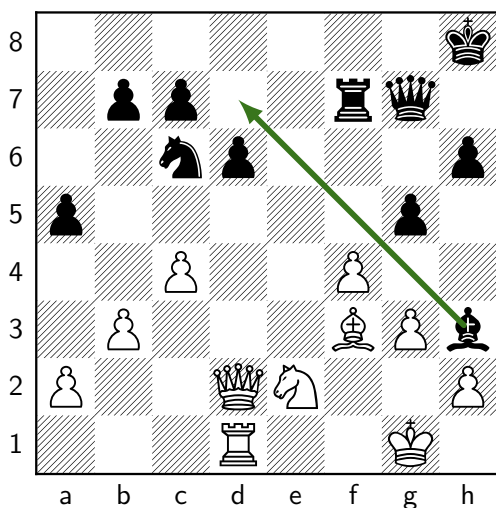
Figure 14: **Move 20, Black: Bxh6.** (a) Our position; Melanie plays Bxh6 (f8h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#612` (ply 49) [structural distance $d=117$, QAP obj=23]: the master played Rxh6; under μ our Bf1 plays the role of that game’s Rh1 (a **cross-type** correspondence); cross-type roles: our Ba6 as their Qc7, our Pe6 as their Be7; transferred move f8h6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#325` (ply 51) [structural distance $d=117$, QAP obj=23]: the master played Rxh6; under μ our Bf1 plays the role of that game’s Rh1 (a **cross-type** correspondence); cross-type roles: our Ba6 as their Qc7, our Pe6 as their Be7; transferred move f8h6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#3481` (ply 46) [structural distance $d=115$, QAP obj=51]: the master played Bxg5; under μ our Bf1 plays the role of that game’s Be3; cross-type roles: our Qd8 as their Rh1, our Bf8 as their Qg3; transferred move f8d6. Synthesis: 2 retrieved analogs transferred to Bxh6 (summed QAP objective 45, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



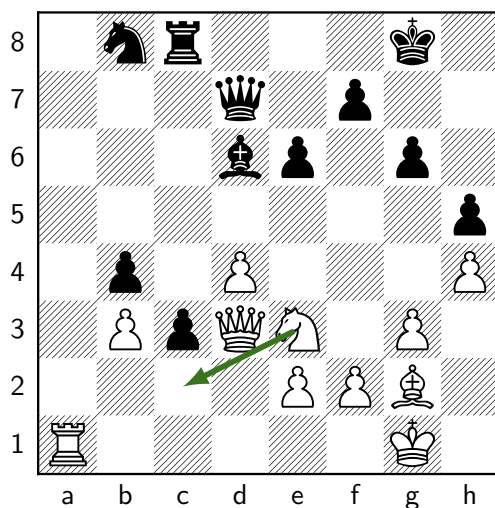
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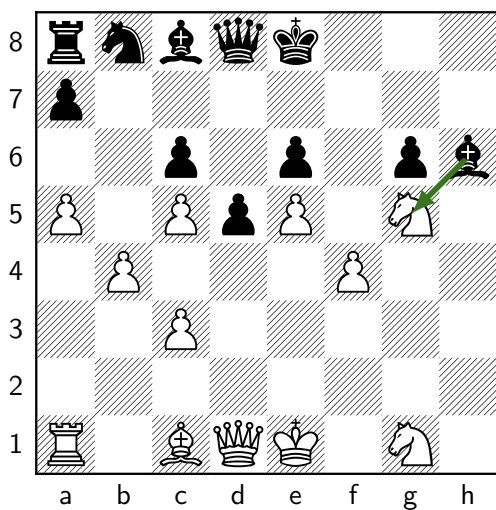


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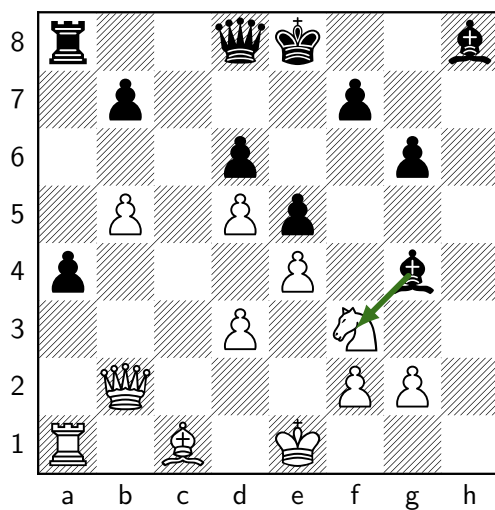


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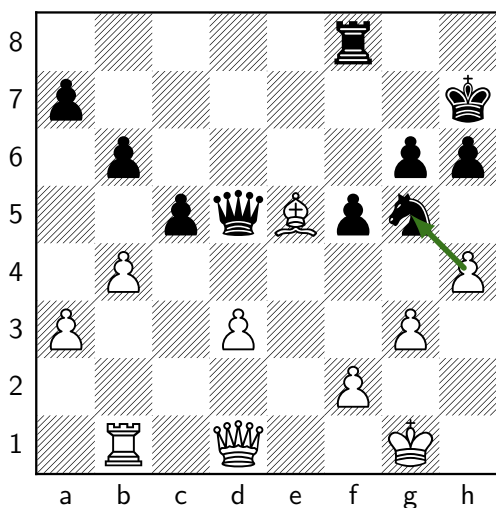
Figure 15: **Move 21, Black: Bc8.** (a) Our position; Melanie plays Bc8 (a6c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2431` (ply 50) [structural distance $d=82$, QAP obj=69]: the master played Bc1; under μ our *Ba3* plays the role of that game’s *Ba3*; cross-type roles: our *Bh6* as their *Re1*, our *Nb8* as their *Pa2*; transferred move a6c8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3808` (ply 57) [structural distance $d=93$, QAP obj=46]: the master played Bd7; under μ our *Ba3* plays the role of that game’s *Bh6*; cross-type roles: our *Bh6* as their *Rf7*, our *pe5* as their *Qg7*; transferred move a6c8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3345` (ply 50) [structural distance $d=87$, QAP obj=24]: the master played Nc2; under μ our *Ba3* plays the role of that game’s *Ne3* (a **cross-type** correspondence); cross-type roles: our *Pd5* as their *Ra1*, our *Ra8* as their *Bg2*; transferred move a6c8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bc8 (summed QAP objective 165, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



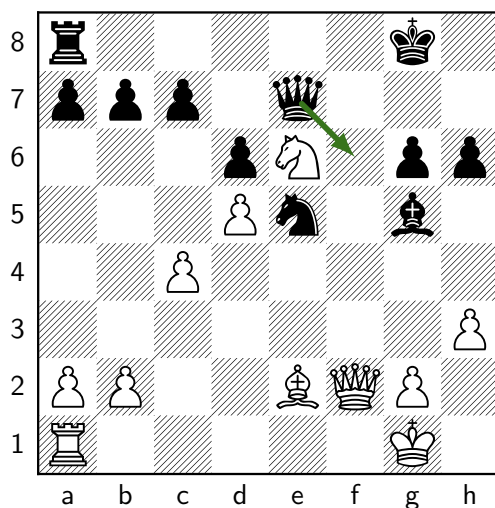
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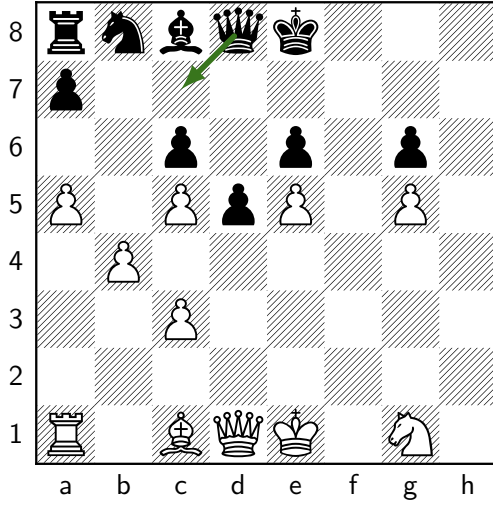


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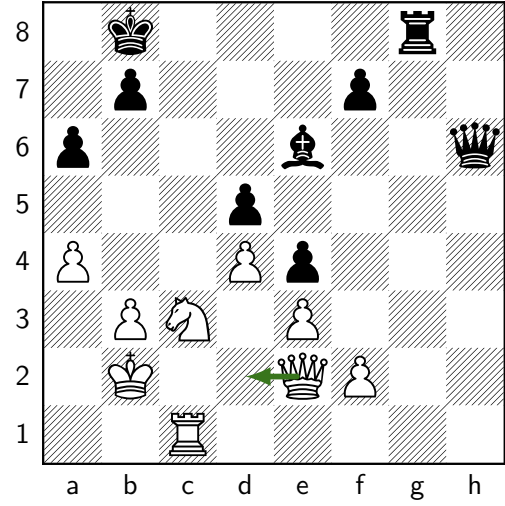


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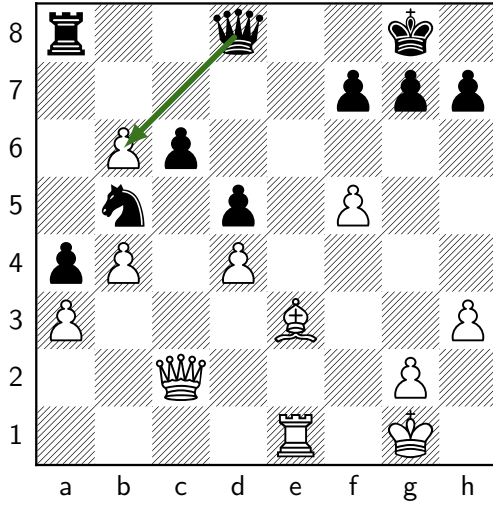
Figure 16: **Move 22, Black: Bxg5.** (a) Our position; Melanie plays Bxg5 (h6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#331` (ply 35) [structural distance $d=61$, QAP obj=48]: the master played Bxf3; under μ our Bh3 plays the role of that game’s Bg5; cross-type roles: our Bc8 as their Ra8, our Ra8 as their Bh8; transferred move h6g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1225` (ply 54) [structural distance $d=81$, QAP obj=35]: the master played hxg5; under μ our Bh3 plays the role of that game’s Ph4 (a **cross-type** correspondence); cross-type roles: our Pe6 as their Rb1, our bc1 as their Pg3; transferred move h6g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2279` (ply 45) [structural distance $d=76$, QAP obj=30]: the master played Qf6; under μ our Bh3 plays the role of that game’s Qe2 (a **cross-type** correspondence); cross-type roles: our Pd5 as their Ra8, our bc1 as their Pb7; transferred move h6g5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bxg5 (summed QAP objective 162, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



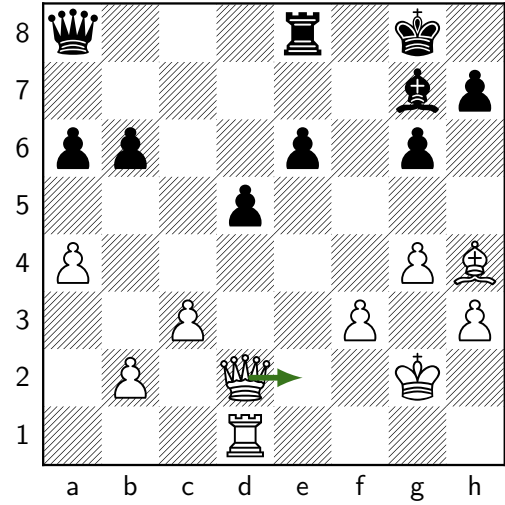
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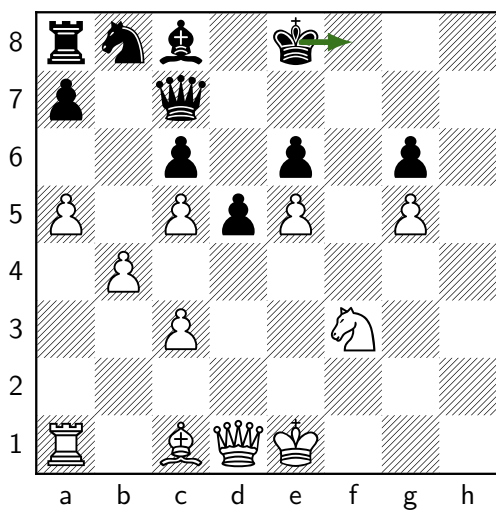


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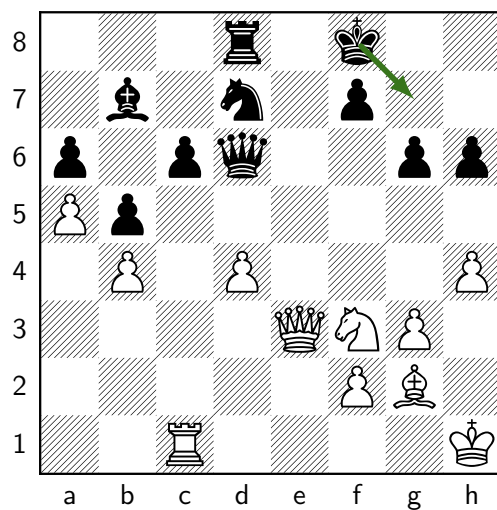


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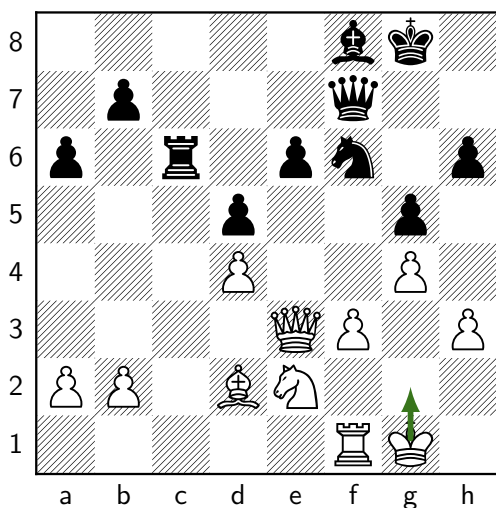
Figure 17: **Move 23, Black: Qc7**. (a) Our position; Melanie plays Qc7 (d8c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2594` (ply 52) [structural distance $d=58$, QAP obj=63]: the master played Qd2; under μ our Qd1 plays the role of that game’s Qe2; cross-type roles: our Pe6 as their Rc1, our Qd8 as their Nc3; transferred move d8c7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1875` (ply 51) [structural distance $d=60$, QAP obj=49]: the master played Qxb6; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Bc8 as their Ra8, our Nb8 as their Pf7; transferred move d8c7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1182` (ply 56) [structural distance $d=60$, QAP obj=42]: the master played Qe2; under μ our Qd1 plays the role of that game’s Qd2; cross-type roles: our Bc8 as their Rd1, our Nb8 as their Ph3; transferred move d8c7 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Qc7 (summed QAP objective 360, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



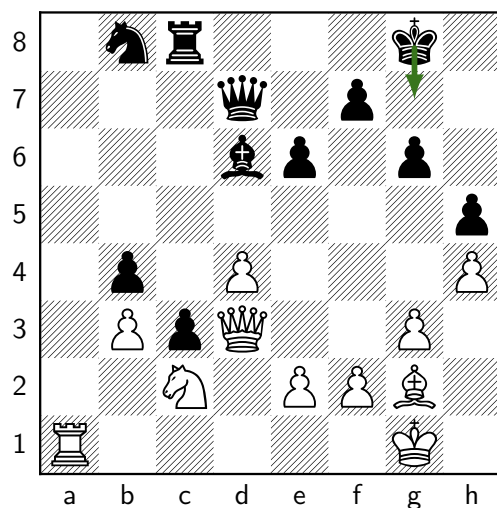
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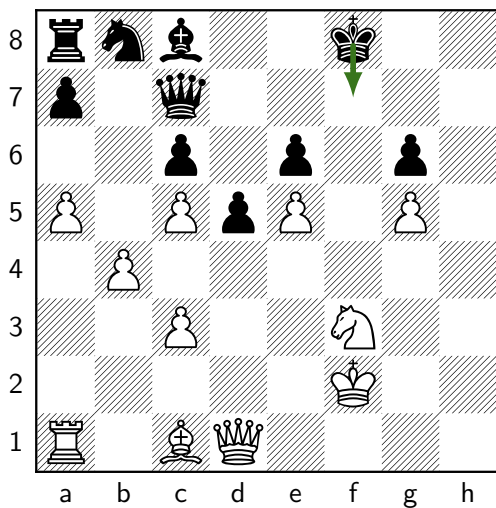


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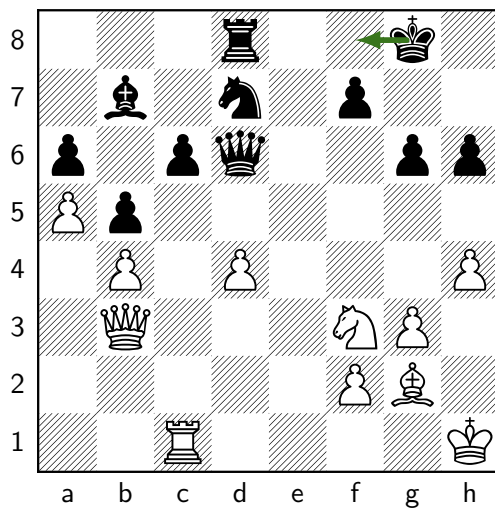


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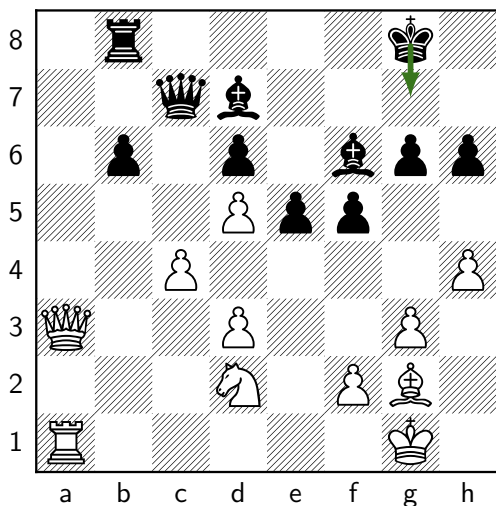
Figure 18: **Move 24, Black: Kf8.** (a) Our position; Melanie plays Kf8 (e8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#525 (ply 57) [structural distance $d=68$, QAP obj=69]: the master played Kg7; under μ our Ke1 plays the role of that game's Kf1; cross-type roles: our Pc6 as their Bb7, our Pa7 as their Nd7; transferred move e8f8 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#715 (ply 52) [structural distance $d=71$, QAP obj=55]: the master played Kg2; under μ our Ke1 plays the role of that game's Kg1; cross-type roles: our Ra8 as their Pa2, our Nb8 as their Bd2; transferred move e8f8 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#3345 (ply 51) [structural distance $d=68$, QAP obj=30]: the master played Kg7; under μ our Ke1 plays the role of that game's Kg1; cross-type roles: our pg5 as their Nb8, our nf3 as their Qd7; transferred move e8f8 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kf8 (summed QAP objective 213, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



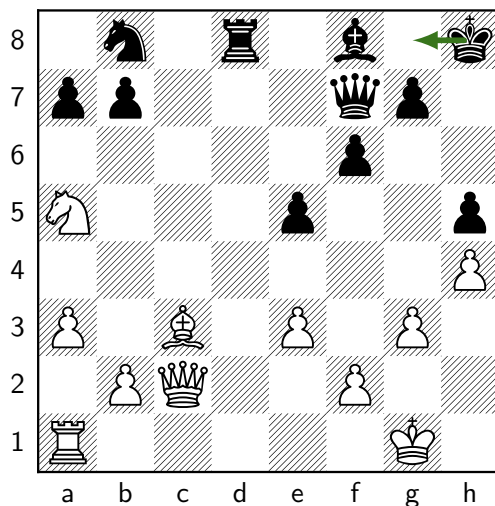
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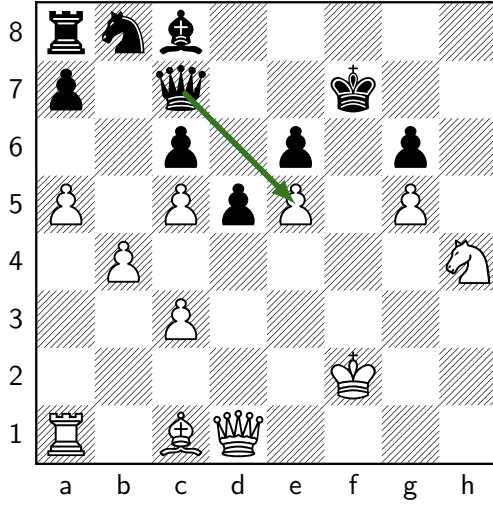


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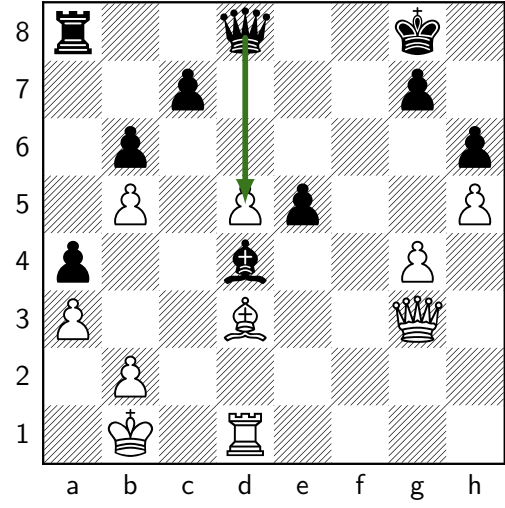


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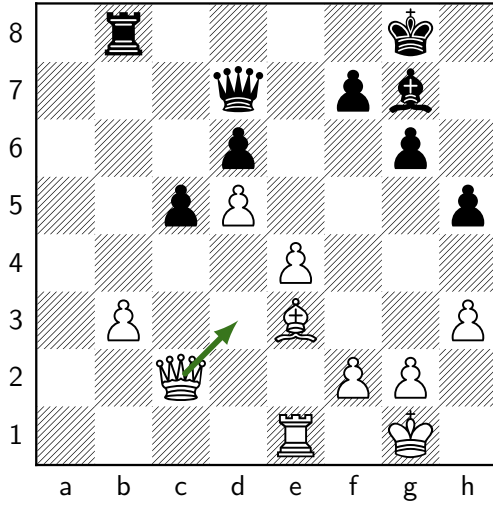
Figure 19: **Move 25, Black: Kf7 (f8f7).** (a) Our position; Melanie plays Kf7 (f8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#525` (ply 55) [structural distance $d=56$, QAP obj=66]: the master played Kf8; under μ our *Kf1* plays the role of that game’s *Kg1*; cross-type roles: our *pb4* as their *Rd8*, our *pc3* as their *Bb7*; transferred move **f8f7 (endorses this move)**. (c) Retrieved from `ccr1_top_p0001#1559` (ply 49) [structural distance $d=54$, QAP obj=42]: the master played Kg7; under μ our *Kf1* plays the role of that game’s *Kg1*; cross-type roles: our *nf3* as their *Rb8*, our *qc7* as their *Bf6*; transferred move **f8f7 (endorses this move)**. (d) Retrieved from `ccr1_top_p0001#3111` (ply 61) [structural distance $d=59$, QAP obj=34]: the master played Kg8; under μ our *Kf1* plays the role of that game’s *Kh1*; cross-type roles: our *Ra8* as their *Nb8*, our *Nb8* as their *Rd8*; transferred move **f8f7 (endorses this move)**. Synthesis: 5 retrieved analogs transferred to Kf7 (summed QAP objective 179, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



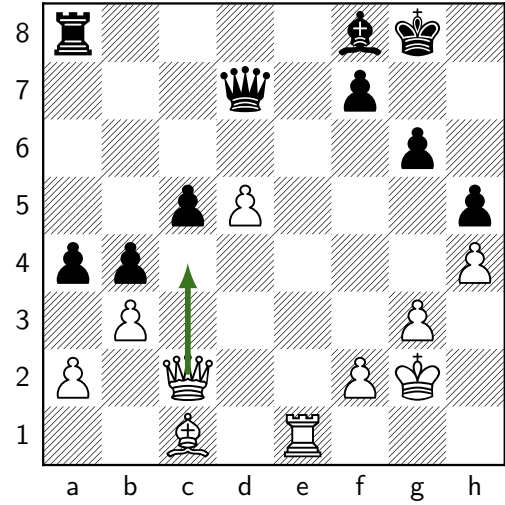
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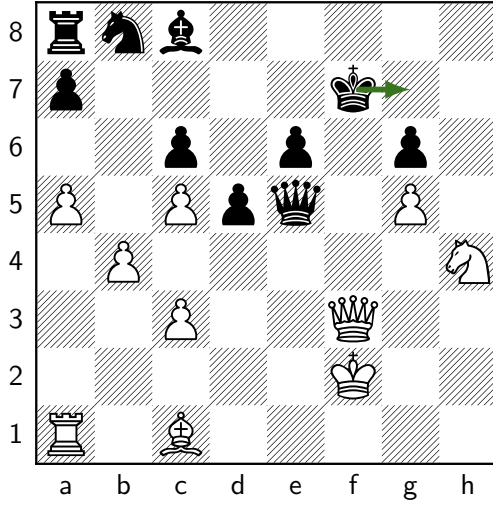


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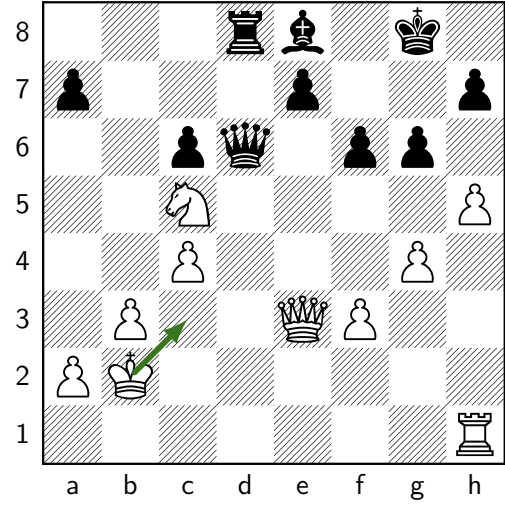


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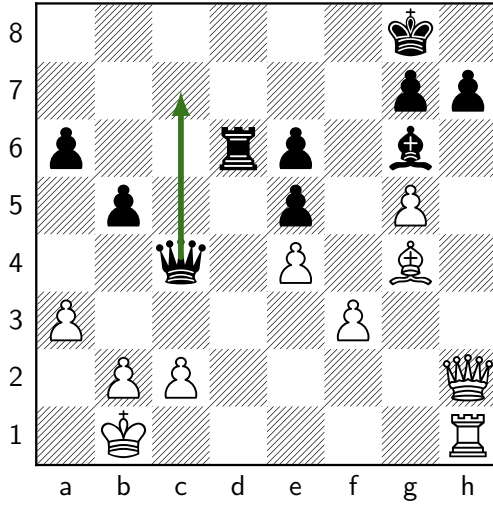
Figure 20: **Move 26, Black: Qxe5**. (a) Our position; Melanie plays Qxe5 (c7e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#2889 (ply 55) [structural distance $d=72$, QAP obj=50]: the master played Qxd5; under μ our Qc2 plays the role of that game’s Qd1; cross-type roles: our Pc5 as their Ra8, our qd1 as their Pc7; transferred move c7e5 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#1282 (ply 54) [structural distance $d=80$, QAP obj=47]: the master played Qd3; under μ our Qc2 plays the role of that game’s Qc2; cross-type roles: our Pc6 as their Re1, our Nb8 as their Pf2; transferred move c7e5 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#1086 (ply 64) [structural distance $d=72$, QAP obj=34]: the master played Qc4; under μ our Qc2 plays the role of that game’s Qc2; cross-type roles: our Nb8 as their Bc1, our Pc6 as their Re1; transferred move c7e5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qxe5 (summed QAP objective 131, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



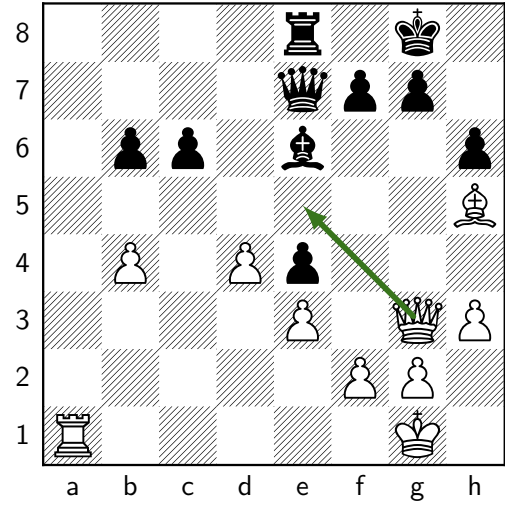
(a)



(b)

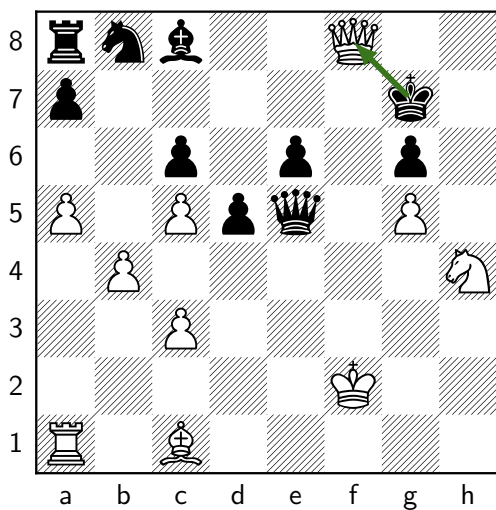


(c)

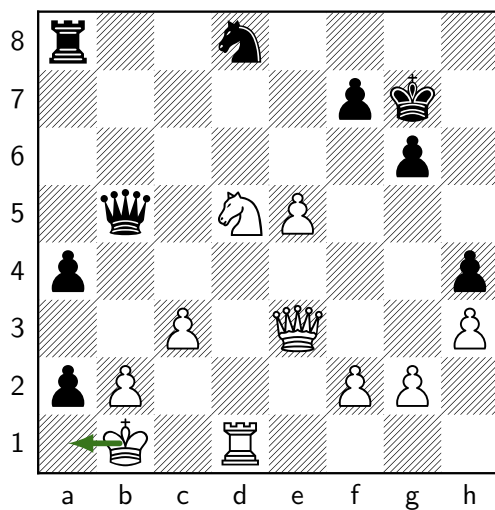


(d)

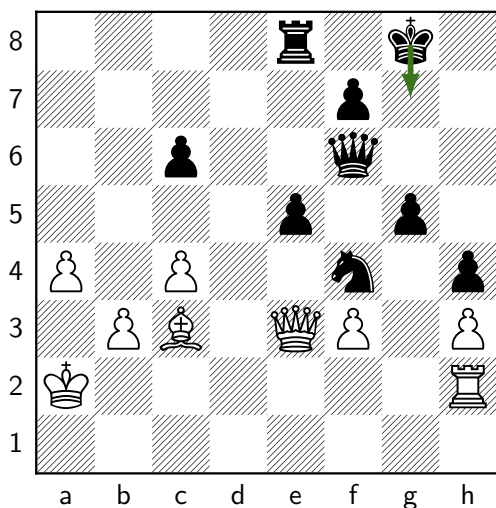
Figure 21: **Move 27, Black: Kg7.** (a) Our position; Melanie plays Kg7 (f7g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#340` (ply 56) [structural distance $d=66$, QAP obj=15]: the master played Kc3; under μ our Kf2 plays the role of that game’s Kb2; cross-type roles: our Nb8 as their Rh1, our qf3 as their Pg4; transferred move f7g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1314` (ply 57) [structural distance $d=64$, QAP obj=34]: the master played Qc7; under μ our Qe4 plays the role of that game’s Qc5; cross-type roles: our nh4 as their Pg7, our Ra8 as their Pa6; transferred move e5f6. (d) Retrieved from `ccr1_top_p0001#2516` (ply 64) [structural distance $d=65$, QAP obj=31]: the master played Qe5; under μ our Qe4 plays the role of that game’s Qg3; cross-type roles: our Pg6 as their Ra1, our Nb8 as their Ph3; transferred move e5f4. Synthesis: 1 retrieved analog transferred to Kg7 (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



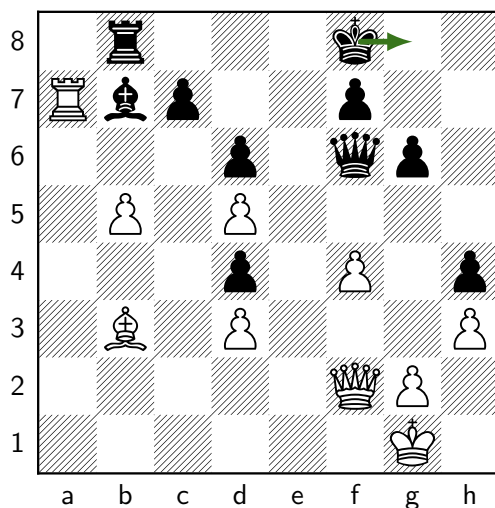
(a)



(b)

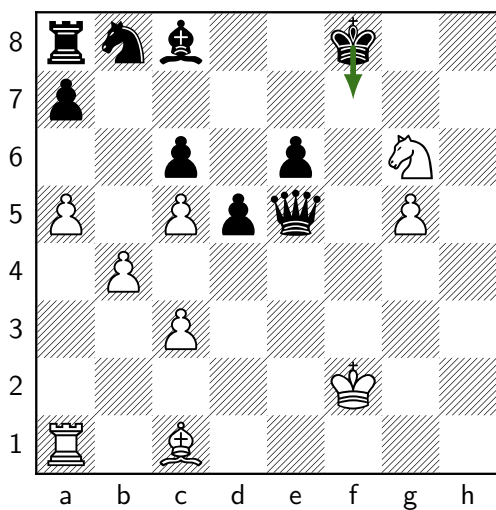


(c)

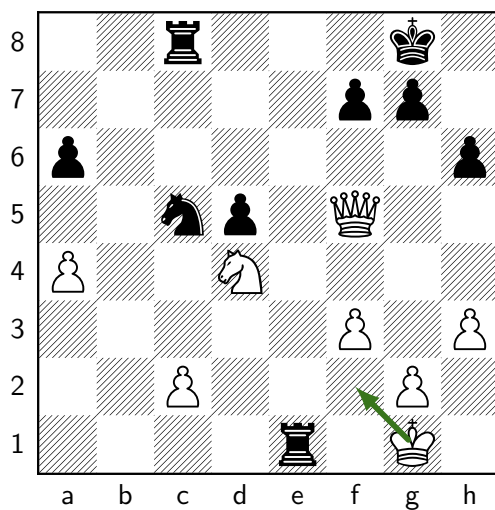


(d)

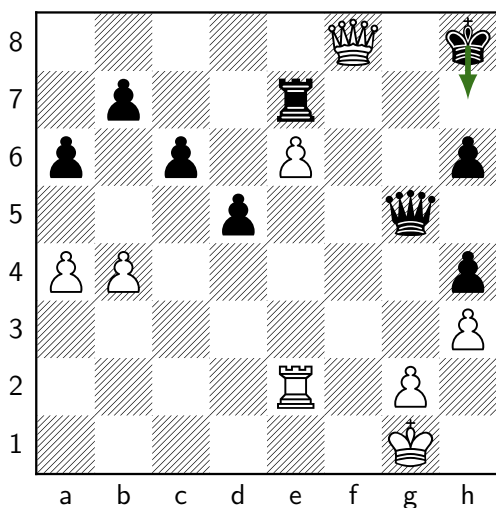
Figure 22: **Move 28, Black: Kxf8.** (a) Our position; Melanie plays Kxf8 (g7f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3001` (ply 58) [structural distance $d=70$, QAP obj=88]: the master played $Ka1$; under μ our $Kg2$ plays the role of that game’s $Kb1$; cross-type roles: our $bc1$ as their $Rd1$, our $qf8$ as their $pa2$; transferred move $g7h7$. (c) Retrieved from `ccrl_top_p0002#1410` (ply 69) [structural distance $d=65$, QAP obj=42]: the master played $Kg7$; under μ our $Kg2$ plays the role of that game’s $Kg1$; cross-type roles: our $bc1$ as their $Qf6$, our $nh4$ as their $pa4$; transferred move $g7h7$. (d) Retrieved from `ccrl_top_p0003#161` (ply 61) [structural distance $d=70$, QAP obj=26]: the master played $Kg8$; under μ our $Kg2$ plays the role of that game’s $Kf1$; cross-type roles: our $Pg6$ as their $ra7$, our $nh4$ as their $Bb7$; transferred move $g7h7$. Synthesis: no retrieved analog transferred to Kxf8 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



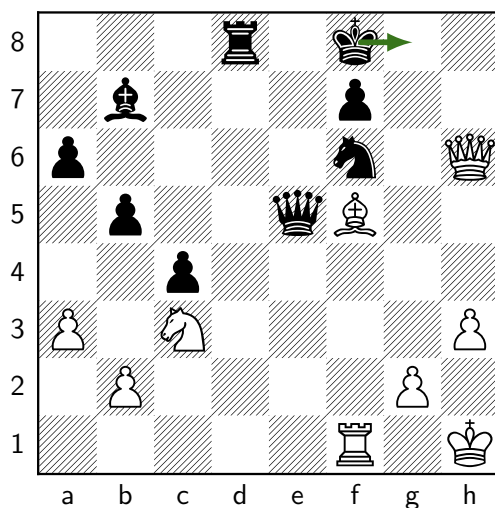
(a)



(b)

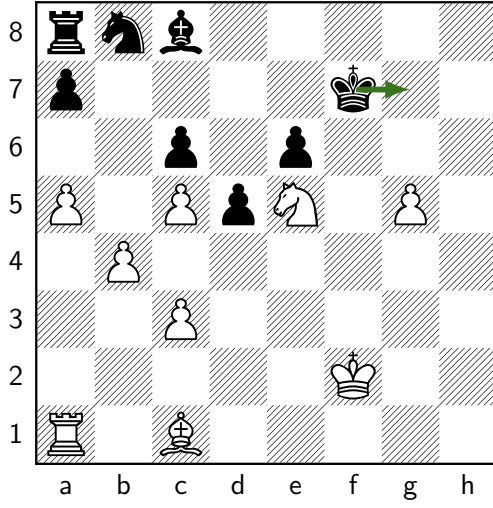


(c)

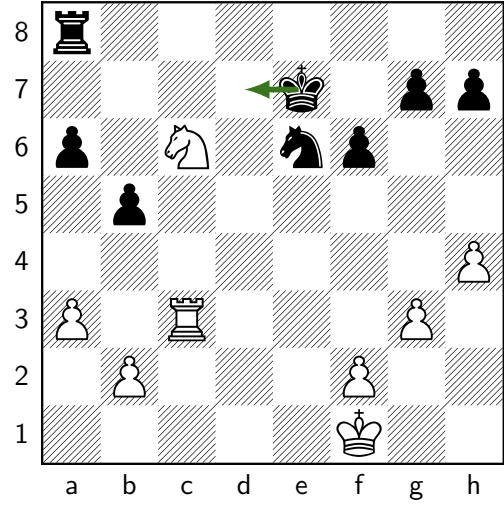


(d)

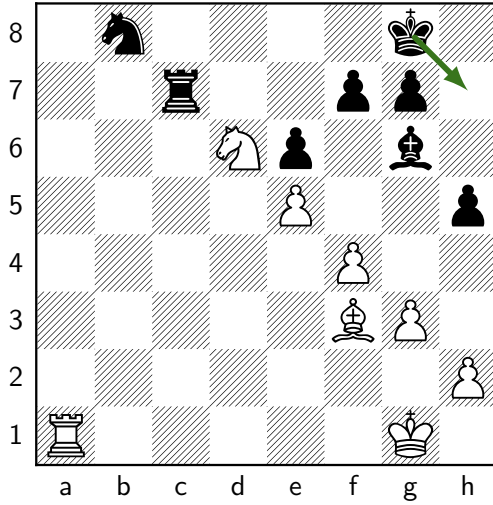
Figure 23: **Move 29, Black: Kf7**. (a) Our position; Melanie plays Kf7 (f8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#864` (ply 64) [structural distance $d=74$, QAP obj=97]: the master played Kf2; under μ our Kf1 plays the role of that game’s Kg1; cross-type roles: our ng6 as their re1, our Nb8 as their Ph3; transferred move f8f7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1504` (ply 73) [structural distance $d=79$, QAP obj=95]: the master played Kh7; under μ our Kf1 plays the role of that game’s Kh1; cross-type roles: our ng6 as their qf8, our Qe5 as their Re7; transferred move f8f7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1245` (ply 59) [structural distance $d=70$, QAP obj=79]: the master played Kg8; under μ our Kf1 plays the role of that game’s Kf1; cross-type roles: our Pc6 as their Rd8, our Qe5 as their Bb7; transferred move f8f7 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Kf7 (summed QAP objective 583, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



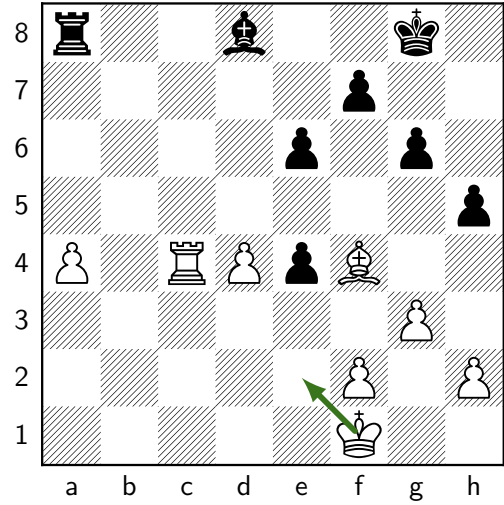
(a)



(b)

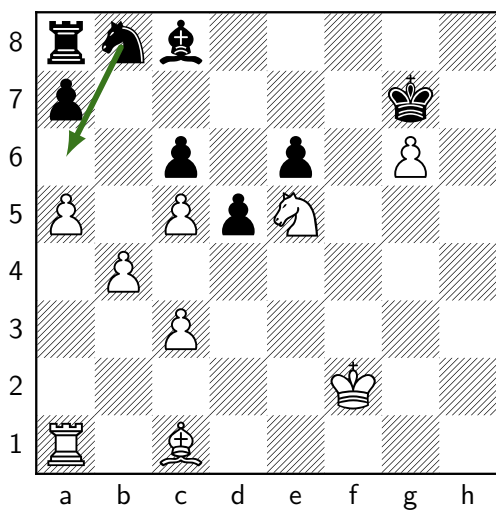


(c)

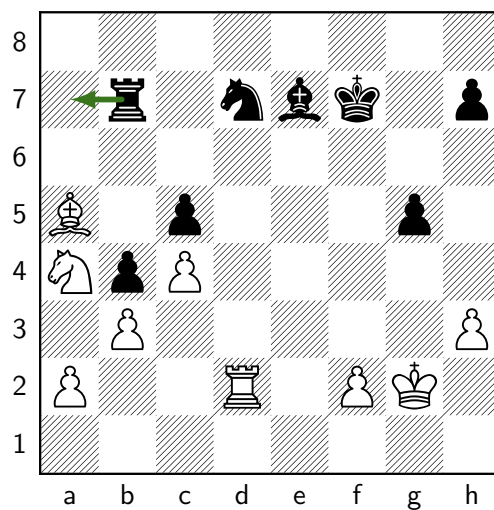


(d)

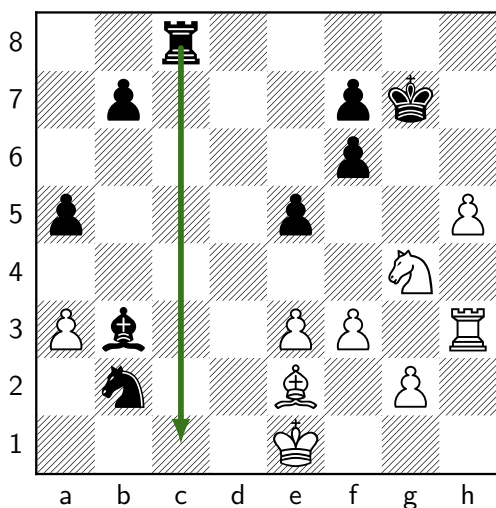
Figure 24: **Move 30, Black: Kg7.** (a) Our position; Melanie plays Kg7 (f7g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3338` (ply 61) [structural distance $d=36$, QAP obj=71]: the master played Kd7; under μ our Kf2 plays the role of that game’s Ke2; cross-type roles: our Nb8 as their Ra8, our Bc8 as their Ph7; transferred move f7g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#956` (ply 59) [structural distance $d=40$, QAP obj=6]: the master played Kh7; under μ our Kf2 plays the role of that game’s Kg1; cross-type roles: our Ra8 as their Nb8, our Nb8 as their Rc7; transferred move f7f8. (d) Retrieved from `ccr1_top_p0002#2564` (ply 48) [structural distance $d=41$, QAP obj=5]: the master played Ke2; under μ our Kf2 plays the role of that game’s Kf1; cross-type roles: our Ra8 as their Pa4, our pg5 as their Rc4; transferred move f7f8. Synthesis: 1 retrieved analog transferred to Kg7 (summed QAP objective 71, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



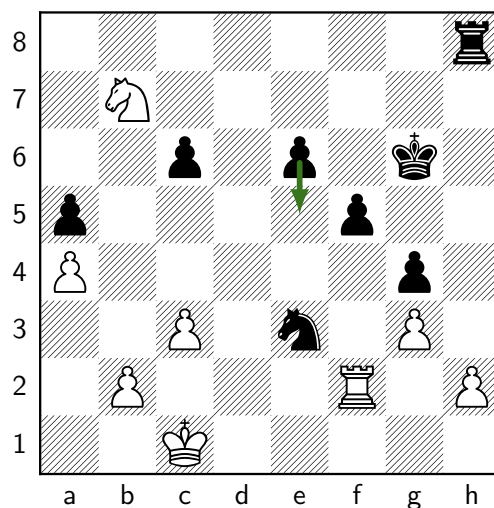
(a)



(b)

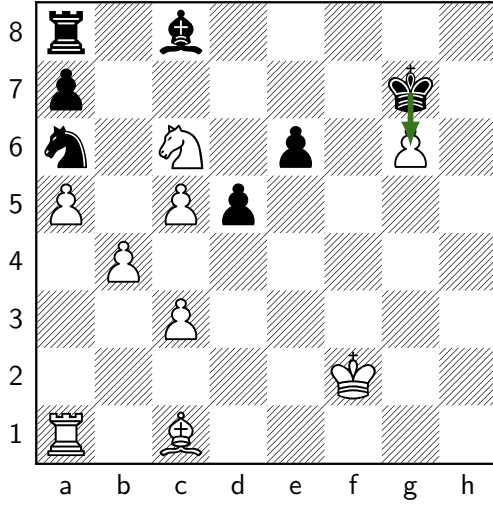


(c)

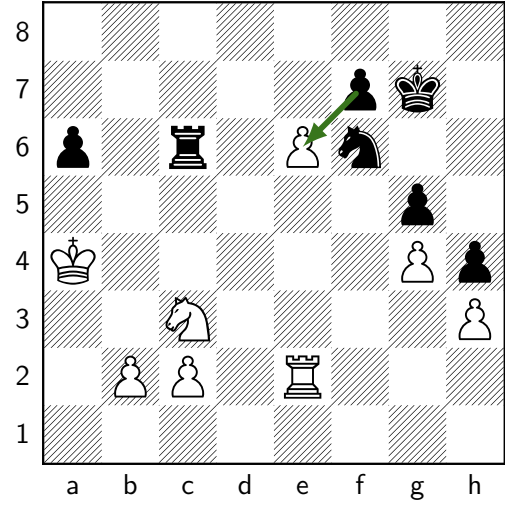


(d)

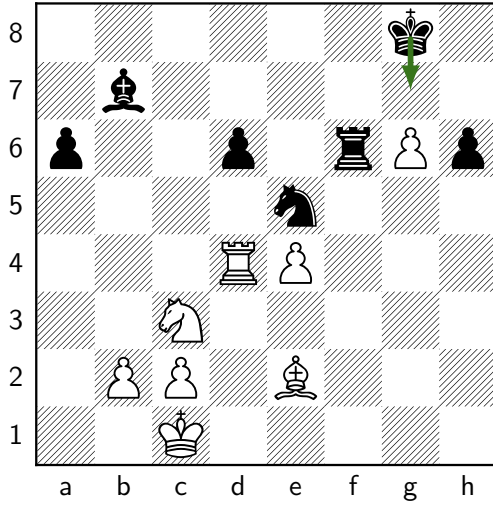
Figure 25: **Move 31, Black: Na6**. (a) Our position; Melanie plays Na6 (b8a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2266` (ply 61) [structural distance $d=43$, QAP obj=25]: the master played Ra7; under μ our Nb1 plays the role of that game's Rb2 (a **cross-type** correspondence); cross-type roles: our Pc6 as their Nd7, our Pa7 as their Be7; transferred move b8a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#380` (ply 57) [structural distance $d=43$, QAP obj=17]: the master played Rc1+; under μ our Nb1 plays the role of that game's Rc1 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Pb7, our pc5 as their Bb3; transferred move b8a6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2326` (ply 73) [structural distance $d=43$, QAP obj=14]: the master played e5; under μ our Nb1 plays the role of that game's Pe3 (a **cross-type** correspondence); cross-type roles: our Kg7 as their nb7, our Bc8 as their pa4; transferred move b8a6 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Na6 (summed QAP objective 96, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



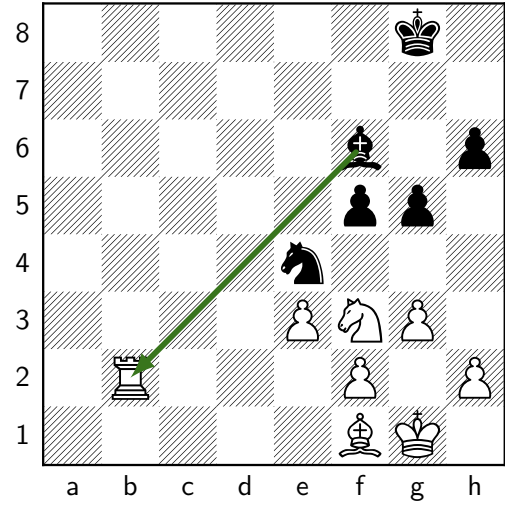
(a)



(b)

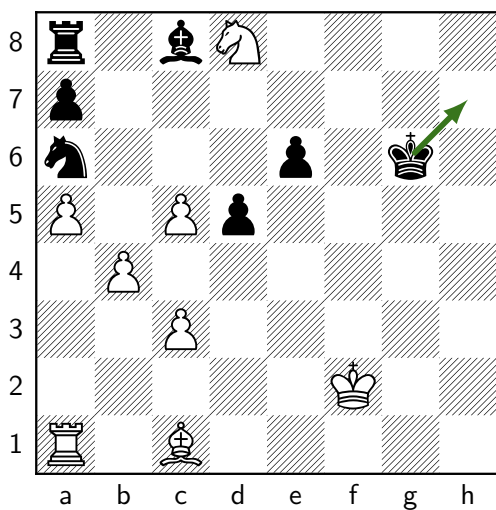


(c)

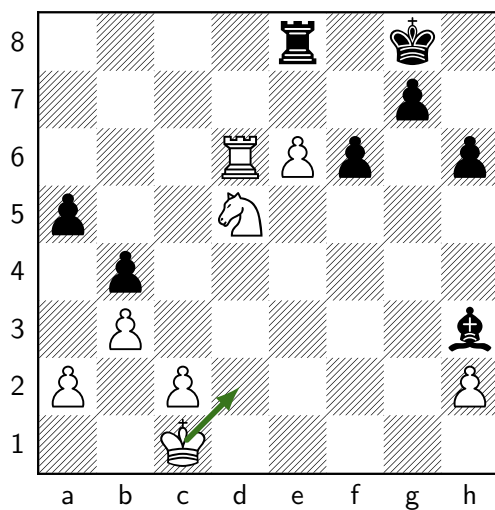


(d)

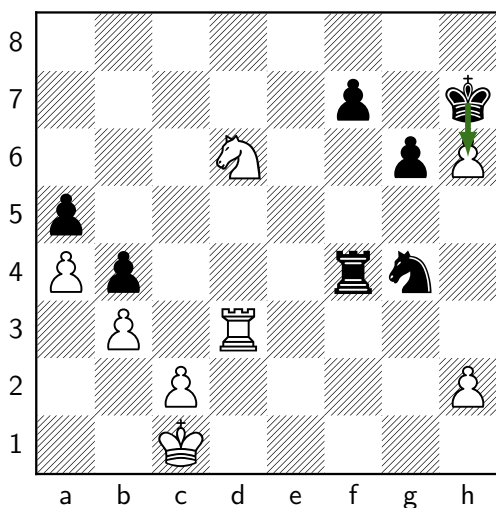
Figure 26: **Move 32, Black: Kxg6**. (a) Our position; Melanie plays Kxg6 (g7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1657` (ply 71) [structural distance $d=58$, QAP obj=42]: the master played fxe6; under μ our Kg2 plays the role of that game’s Pf2 (a **cross-type** correspondence); cross-type roles: our Kg7 as their Rc6, our pg6 as their nc3; transferred move g7g6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2483` (ply 53) [structural distance $d=55$, QAP obj=39]: the master played Kg7; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our Na6 as their Bb7, our Kg7 as their Rf6; transferred move g7g6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#311` (ply 67) [structural distance $d=57$, QAP obj=36]: the master played Bxb2; under μ our Kg2 plays the role of that game’s Bf3 (a **cross-type** correspondence); cross-type roles: our Ra8 as their Ph6, our pg6 as their rb2; transferred move g7g6 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kxg6 (summed QAP objective 156, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



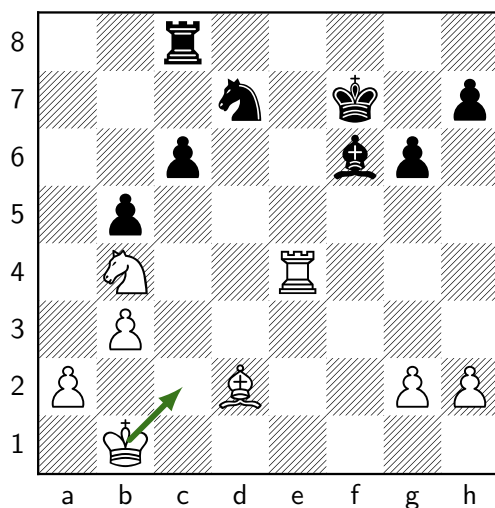
(a)



(b)

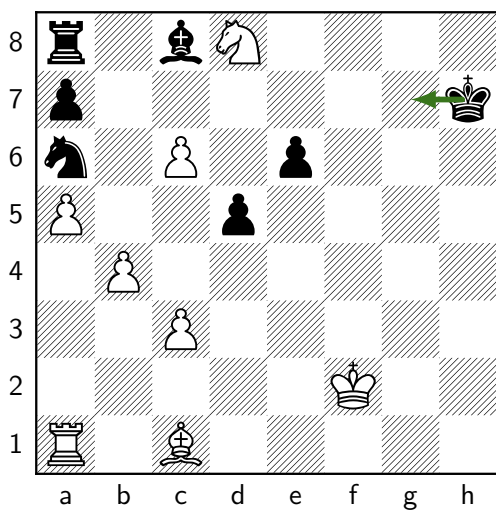


(c)

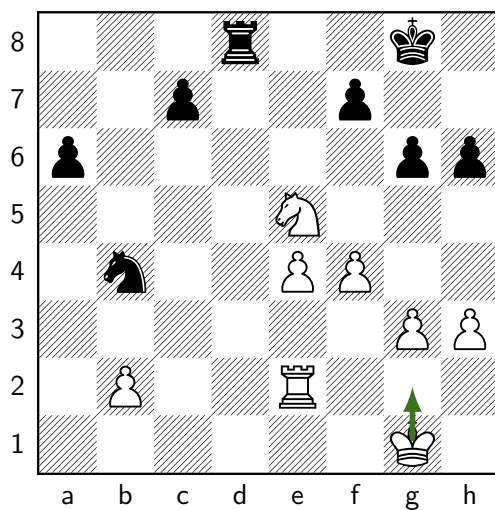


(d)

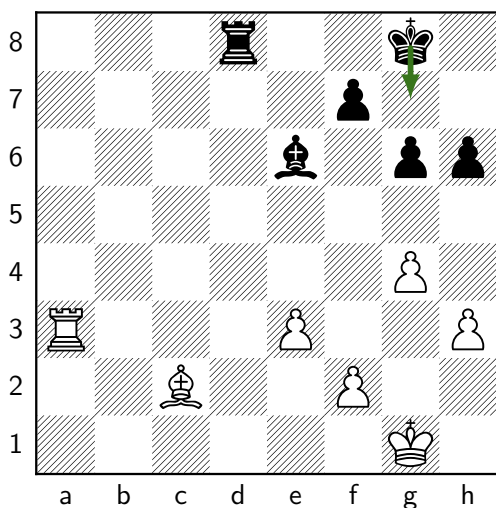
Figure 27: **Move 33, Black: Kh7**. (a) Our position; Melanie plays Kh7 (g6h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#1702 (ply 70) [structural distance $d=44$, QAP obj=74]: the master played Kd2; under μ our Kg3 plays the role of that game’s Kc1; cross-type roles: our Ra8 as their Pa2, our bc1 as their Pc2; transferred move g6h7 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#1082 (ply 77) [structural distance $d=48$, QAP obj=36]: the master played Kxh6; under μ our Kg3 plays the role of that game’s Kh2; cross-type roles: our Ra8 as their Pa5, our Bc8 as their Rf4; transferred move g6h7 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#3300 (ply 74) [structural distance $d=46$, QAP obj=25]: the master played Kc2; under μ our Kg3 plays the role of that game’s Kb1; cross-type roles: our Ra8 as their Pg2, our nd8 as their Ph2; transferred move g6h7 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kh7 (summed QAP objective 179, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



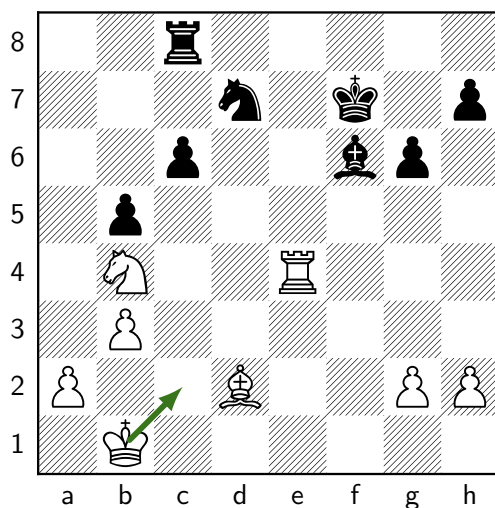
(a)



(b)

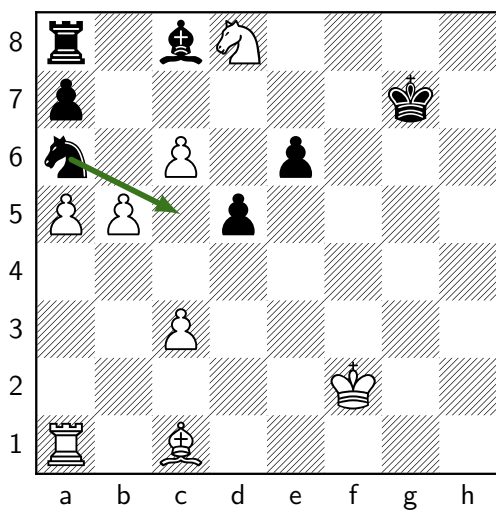


(c)

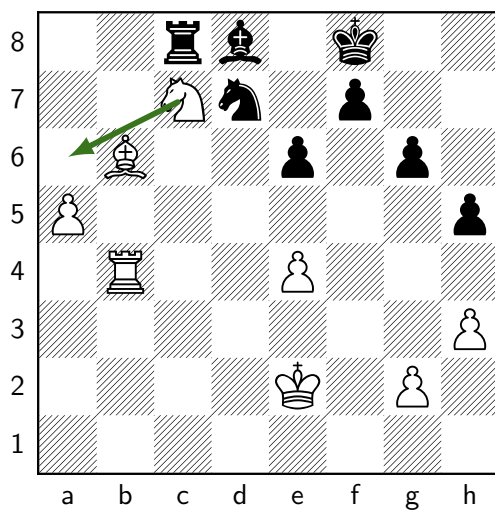


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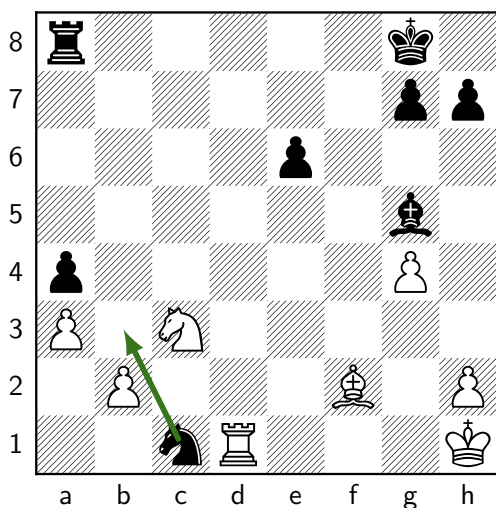
Figure 28: **Move 34, Black: Kg7**. (a) Our position; Melanie plays Kg7 (h7g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#698` (ply 78) [structural distance $d=45$, QAP obj=35]: the master played Kg2; under μ our Kh2 plays the role of that game’s Kg1; cross-type roles: our Ra8 as their Pb2, our Bc8 as their Re2; transferred move h7g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1935` (ply 59) [structural distance $d=45$, QAP obj=30]: the master played Kg7; under μ our Kh2 plays the role of that game’s Kg1; cross-type roles: our pa5 as their Rd8, our nd8 as their Be6; transferred move h7g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3300` (ply 74) [structural distance $d=36$, QAP obj=26]: the master played Kc2; under μ our Kh2 plays the role of that game’s Kb1; cross-type roles: our Ra8 as their Pg2, our nd8 as their Ph2; transferred move h7g7 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kg7 (summed QAP objective 142, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



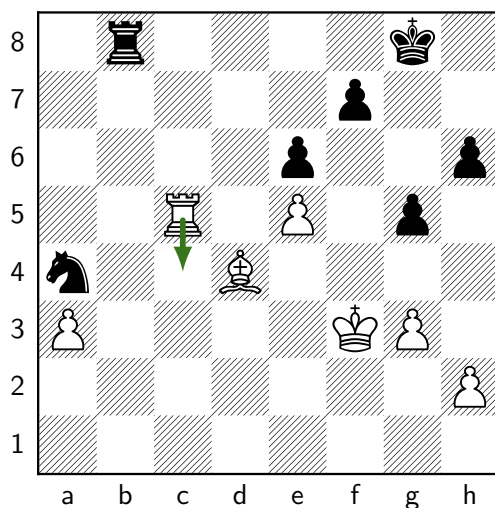
(a)



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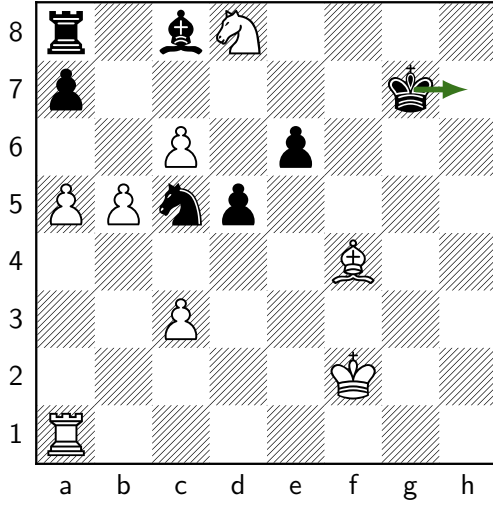


(c)

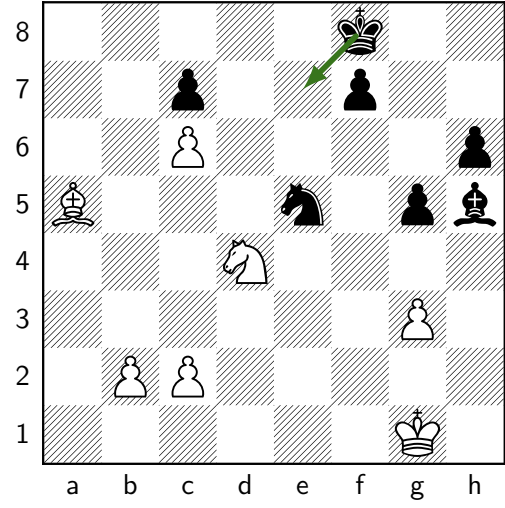


(d)

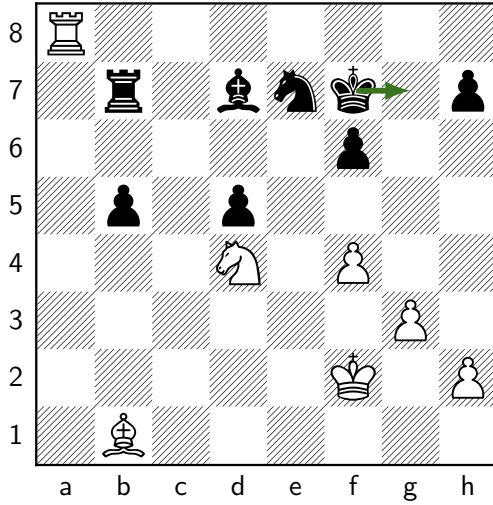
Figure 29: **Move 35, Black: Nc5.** (a) Our position; Melanie plays Nc5 (a6c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2390` (ply 60) [structural distance $d=39$, QAP obj=34]: the master played Na6; under μ our Na3 plays the role of that game’s Nc7; cross-type roles: our Bc8 as their Rb4, our Ra8 as their Pe4; transferred move a6c5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2901` (ply 63) [structural distance $d=43$, QAP obj=20]: the master played Nb3; under μ our Na3 plays the role of that game’s Nc8; cross-type roles: our Pd5 as their Ra8, our Ra8 as their Pg7; transferred move a6c5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2586` (ply 84) [structural distance $d=46$, QAP obj=18]: the master played Rc4; under μ our Na3 plays the role of that game’s Rc5 (a **cross-type** correspondence); cross-type roles: our ra1 as their Ph2, our Ra8 as their Pa3; transferred move a6c5 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Nc5 (summed QAP objective 153, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



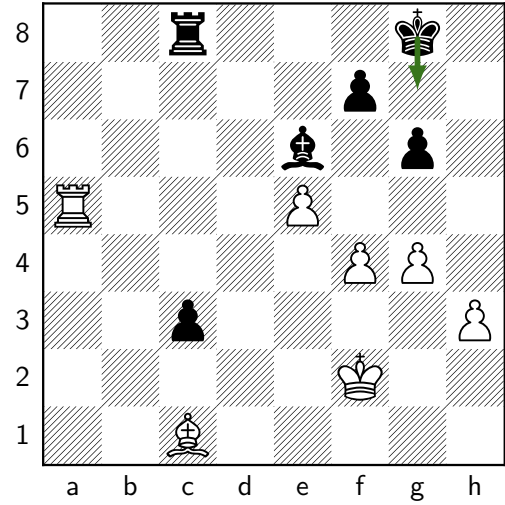
(a)



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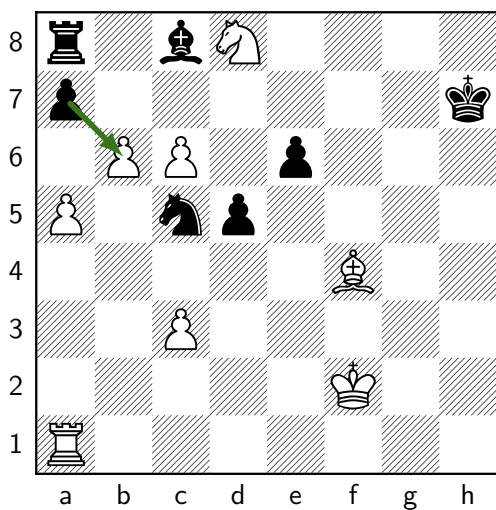


(c)

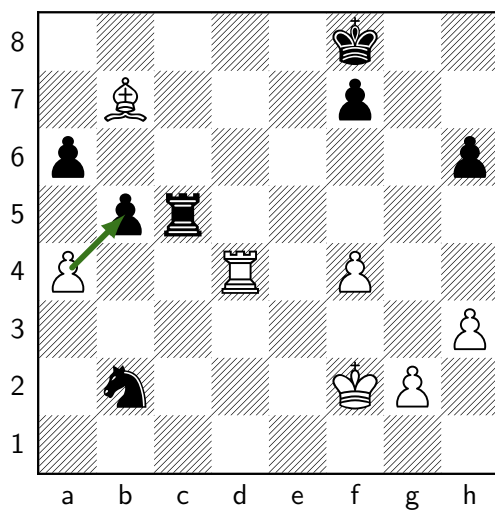


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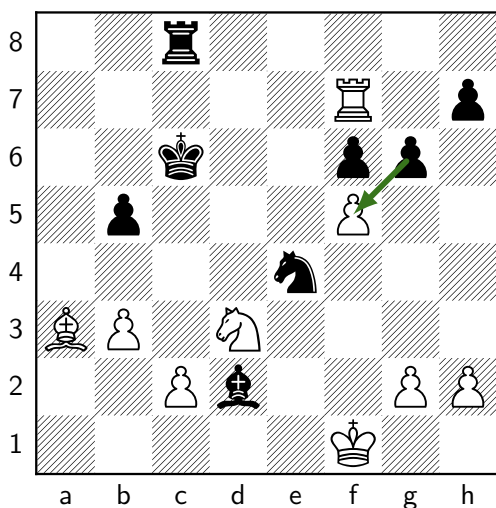
Figure 30: **Move 36, Black: Kh7**. (a) Our position; Melanie plays Kh7 (g7h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3102` (ply 65) [structural distance $d=39$, QAP obj=33]: the master played Ke7; under μ our Kg2 plays the role of that game's Kf1; cross-type roles: our nd8 as their ba5, our pb5 as their Bh5; transferred move g7h7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1750` (ply 73) [structural distance $d=36$, QAP obj=22]: the master played Kg7; under μ our Kg2 plays the role of that game's Kf2; cross-type roles: our pc6 as their ra8, our Bc8 as their Rb7; transferred move g7h7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2533` (ply 57) [structural distance $d=39$, QAP obj=22]: the master played Kg7; under μ our Kg2 plays the role of that game's Kg1; cross-type roles: our pc6 as their Rc8, our nd8 as their Be6; transferred move g7h7 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kh7 (summed QAP objective 114, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



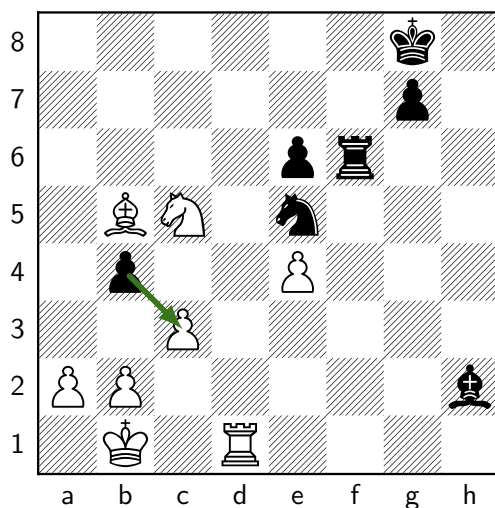
(a)



(b)

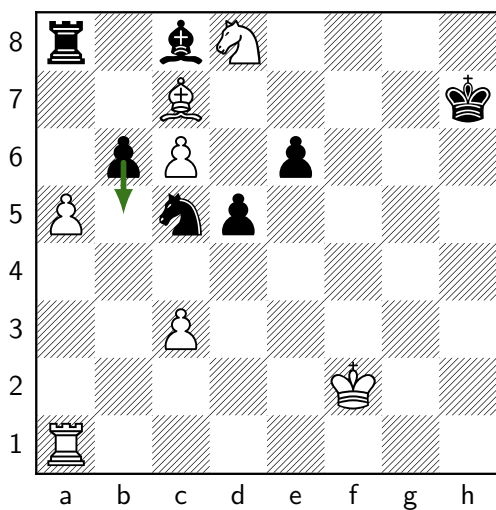


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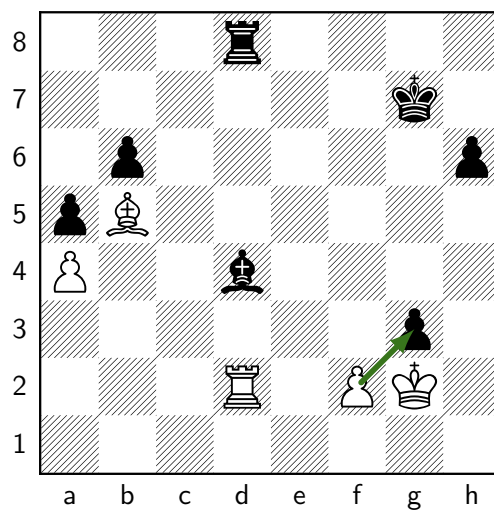


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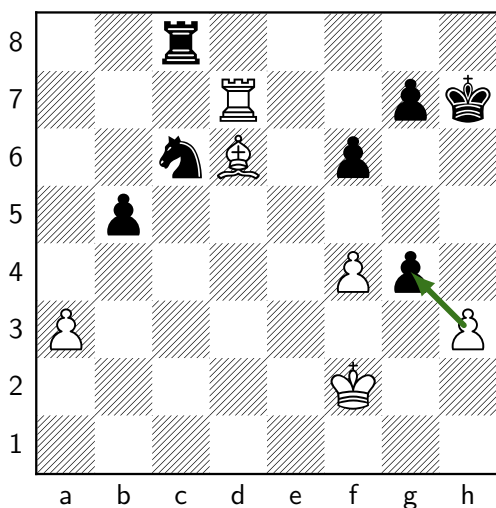
Figure 31: **Move 37, Black: axb6.** (a) Our position; Melanie plays axb6 (a7b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#235 (ply 78) [structural distance $d=40$, QAP obj=58]: the master played axb5; under μ our Pa2 plays the role of that game’s Pa4; cross-type roles: our pb6 as their nb2, our Bc8 as their Pf4; transferred move a7b6 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#423 (ply 69) [structural distance $d=42$, QAP obj=54]: the master played gxf5; under μ our Pa2 plays the role of that game’s Pg3; cross-type roles: our pb6 as their rf7, our Bc8 as their Pb5; transferred move a7b6 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#3711 (ply 65) [structural distance $d=40$, QAP obj=52]: the master played bxc3; under μ our Pa2 plays the role of that game’s Pb5; cross-type roles: our bf4 as their Pg7, our Bc8 as their Rf6; transferred move a7b6 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to axb6 (summed QAP objective 351, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



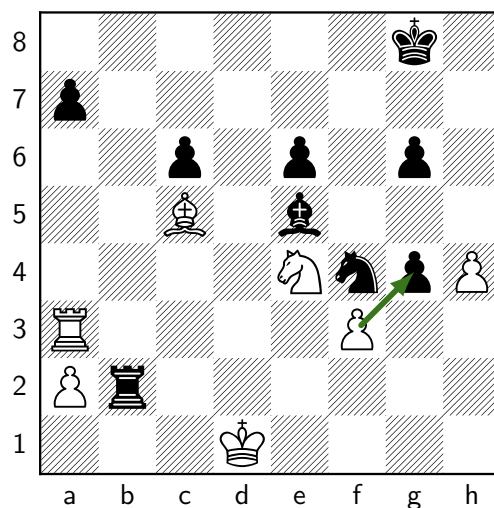
(a)



(b)

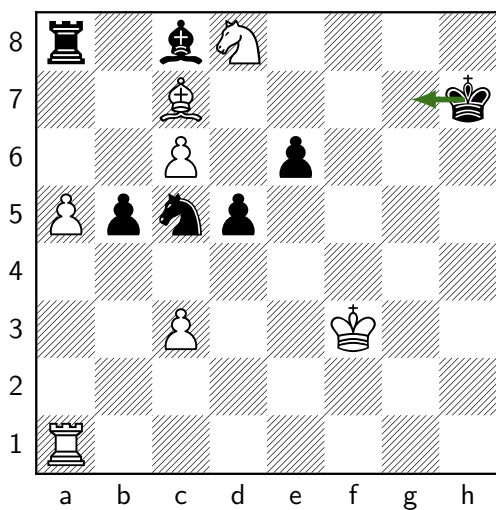


(c)

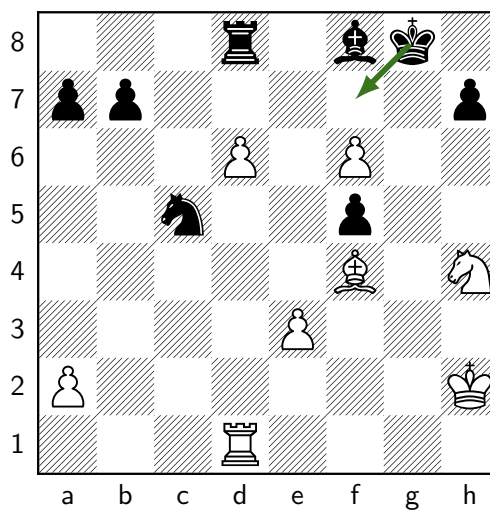


(d)

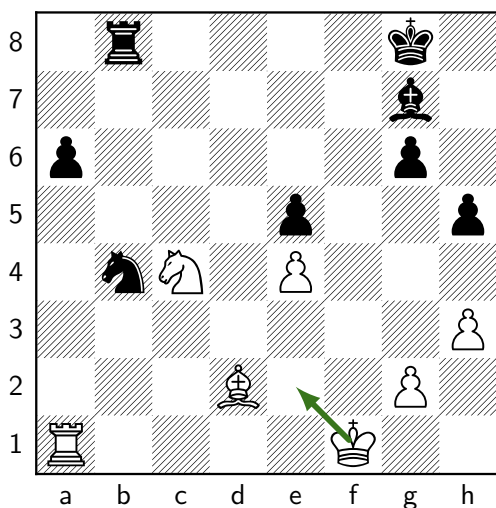
Figure 32: **Move 38, Black: b5 (b6b5)**. (a) Our position; Melanie plays b5 (b6b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2057` (ply 84) [structural distance $d=45$, QAP obj=65]: the master played `fxg3`; under μ our *Pb3* plays the role of that game’s *Pf2*; cross-type roles: our *Bc8* as their *Pa4*, our *Nc5* as their *Bb5*; transferred move `b6b5` (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3998` (ply 74) [structural distance $d=46$, QAP obj=64]: the master played `hxg4`; under μ our *Pb3* plays the role of that game’s *Ph3*; cross-type roles: our *Nc5* as their *Pa3*, our *bc7* as their *rc8*; transferred move `b6b5` (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2580` (ply 78) [structural distance $d=41$, QAP obj=64]: the master played `fxg4`; under μ our *Pb3* plays the role of that game’s *Pf3*; cross-type roles: our *bc7* as their *rb2*, our *ra1* as their *nf4*; transferred move `b6b5` (**endorses this move**). Synthesis: 10 retrieved analogs transferred to b5 (summed QAP objective 475, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



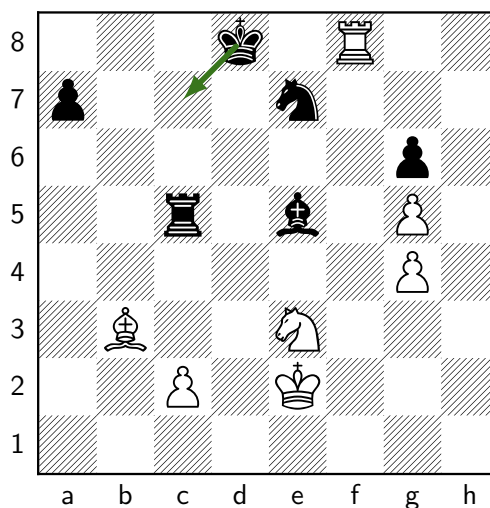
(a)



(b)



(c)



(d)

Figure 33: **Move 39, Black: Kg7.** (a) Our position; Melanie plays Kg7 (h7g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3171` (ply 63) [structural distance $d=39$, QAP obj=55]: the master played Kf7; under μ our Kh2 plays the role of that game’s Kg1; cross-type roles: our Ra8 as their Bf8, our Bc8 as their Pa7; transferred move h7g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#743` (ply 68) [structural distance $d=35$, QAP obj=12]: the master played Ke2; under μ our Kh2 plays the role of that game’s Kf1; cross-type roles: our pc6 as their Ra1, our Ra8 as their Bd2; transferred move h7g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#914` (ply 87) [structural distance $d=35$, QAP obj=4]: the master played Kc7; under μ our Kh2 plays the role of that game’s Kd1; cross-type roles: our Nc5 as their rf8, our Bc8 as their Pa7; transferred move h7g7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kg7 (summed QAP objective 76, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

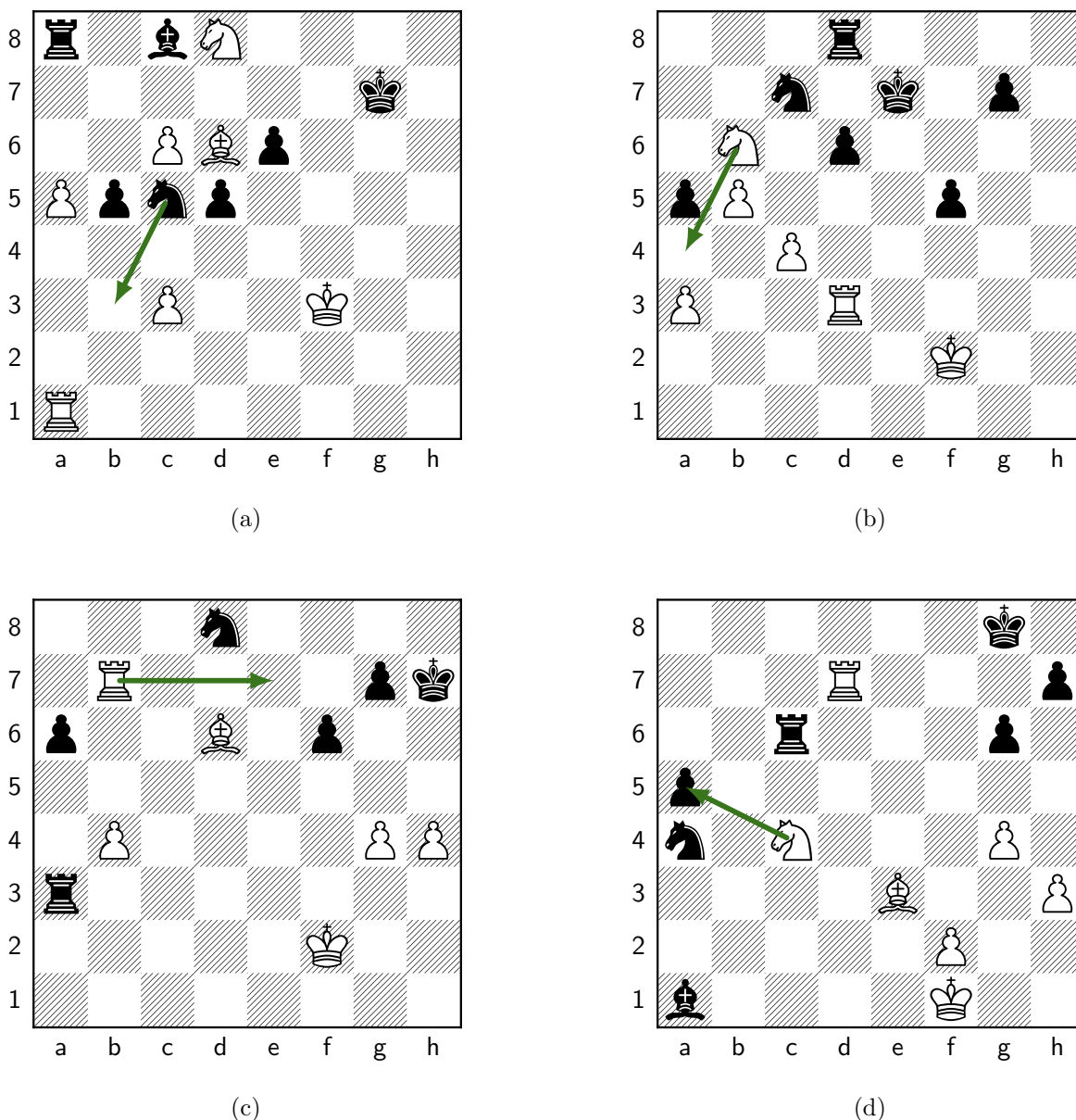


Figure 34: **Move 40, Black: Nb3.** (a) Our position; Melanie plays Nb3 (c5b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#3894 (ply 88) [structural distance $d=32$, QAP obj=26]: the master played Na4; under μ our Nc4 plays the role of that game's Nb6; cross-type roles: our Bc8 as their Pa3, our nd8 as their pf5; transferred move c5b3 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#3464 (ply 80) [structural distance $d=35$, QAP obj=20]: the master played Re7; under μ our Nc4 plays the role of that game's Rb7 (a **cross-type** correspondence); cross-type roles: our nd8 as their ra3, our Bc8 as their Pb4; transferred move c5b3 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#1172 (ply 88) [structural distance $d=26$, QAP obj=13]: the master played Nxa5; under μ our Nc4 plays the role of that game's Nc4; cross-type roles: our nd8 as their ba1, our Pe6 as their Be3; transferred move c5b3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nb3 (summed QAP objective 70, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

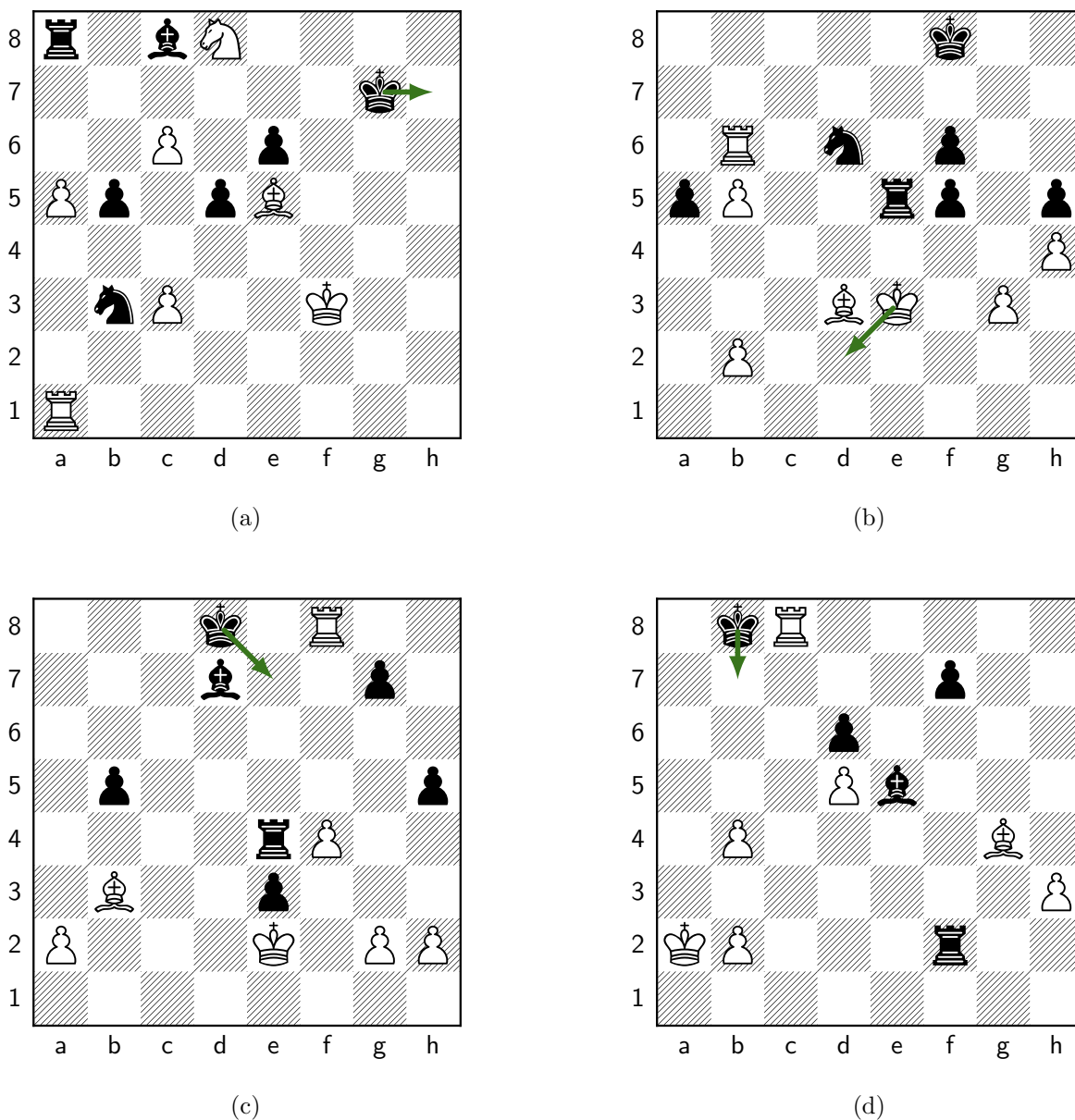
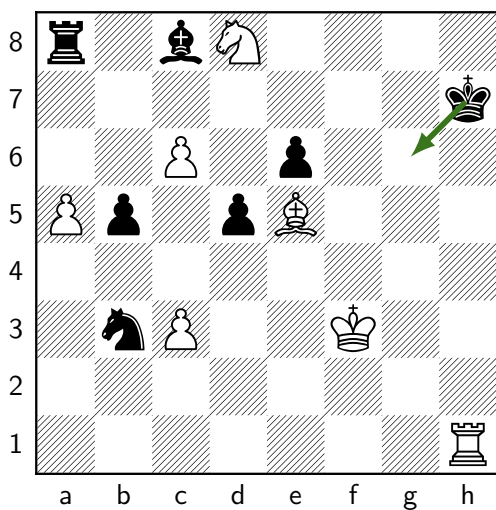
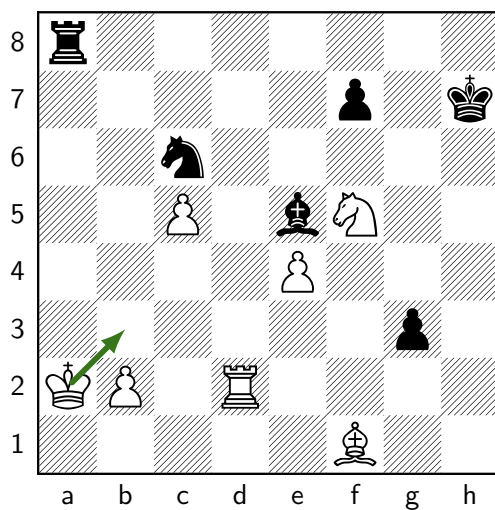


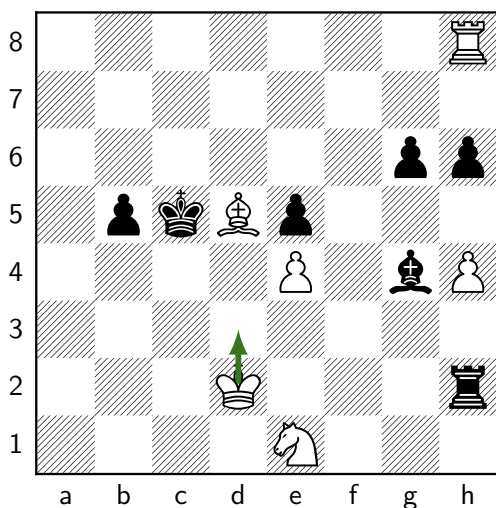
Figure 35: **Move 41, Black: Kh7.** (a) Our position; Melanie plays Kh7 (g7h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#3986 (ply 94) [structural distance $d=39$, QAP obj=69]: the master played Kd2; under μ our Kg2 plays the role of that game's Ke3; cross-type roles: our Ra8 as their Pb2, our Pd5 as their Bd3; transferred move g7h7 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#1750 (ply 71) [structural distance $d=37$, QAP obj=67]: the master played Ke7; under μ our Kg2 plays the role of that game's Kd1; cross-type roles: our be5 as their rf8, our Pd5 as their Bd7; transferred move g7h7 (**endorses this move**). (d) Retrieved from ccr1_top_p0003#1611 (ply 75) [structural distance $d=39$, QAP obj=51]: the master played Kb7; under μ our Kg2 plays the role of that game's Kb1; cross-type roles: our be5 as their rc8, our Bc8 as their Pf7; transferred move g7h7 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kh7 (summed QAP objective 341, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



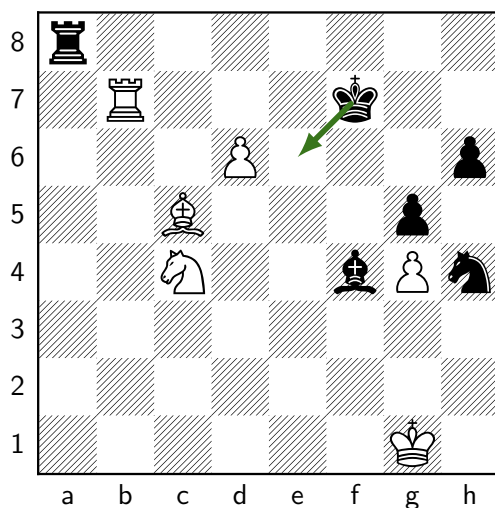
(a)



(b)

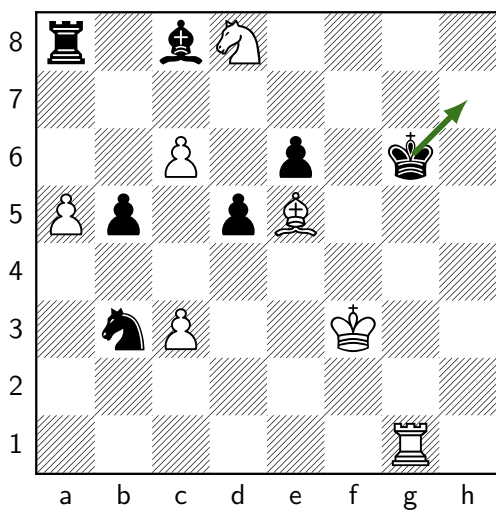


(c)

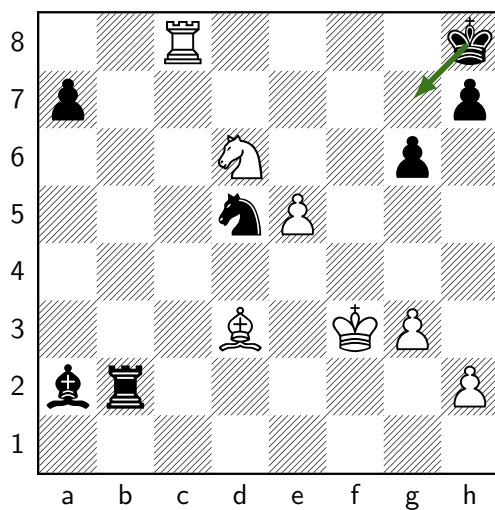


(d)

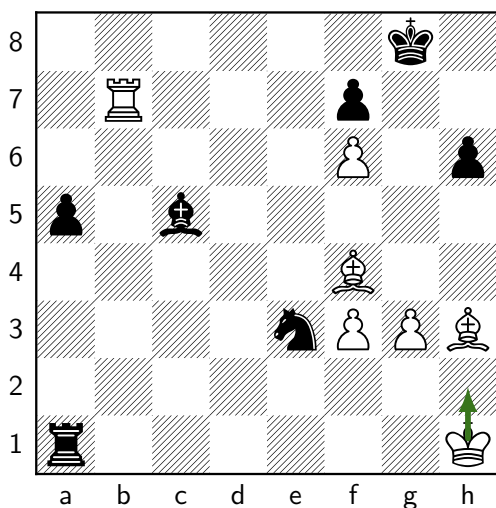
Figure 36: **Move 42, Black: Kg6.** (a) Our position; Melanie plays Kg6 (**h7g6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2862` (ply 92) [structural distance $d=27$, QAP obj=69]: the master played Kb3; under μ our Kh2 plays the role of that game’s Ka2; cross-type roles: our Nb3 as their Rd2, our Ra8 as their nc6; transferred move **h7g6 (endorses this move)**. (c) Retrieved from `ccr1_top_p0001#1845` (ply 90) [structural distance $d=30$, QAP obj=48]: the master played Kd3; under μ our Kh2 plays the role of that game’s Kd2; cross-type roles: our Pe6 as their Bd5, our Bc8 as their pg6; transferred move **h7g6 (endorses this move)**. (d) Retrieved from `ccr1_top_p0003#1209` (ply 95) [structural distance $d=31$, QAP obj=48]: the master played Ke6; under μ our Kh2 plays the role of that game’s Kf2; cross-type roles: our Nb3 as their Ra8, our nd8 as their Ph6; transferred move **h7g6 (endorses this move)**. Synthesis: 15 retrieved analogs transferred to Kg6 (summed QAP objective 346, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



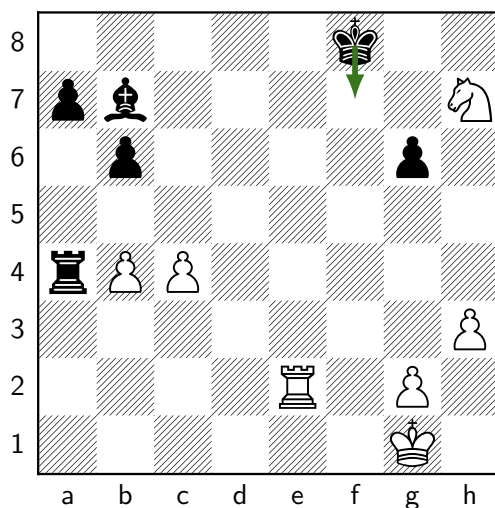
(a)



(b)

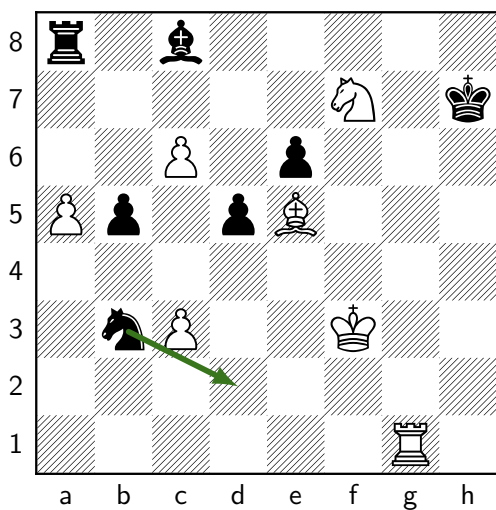


(c)

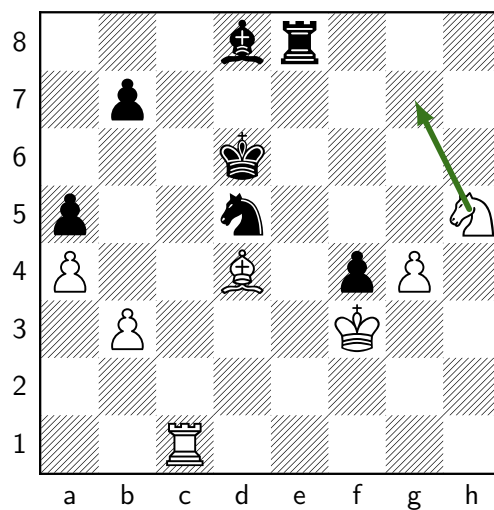


(d)

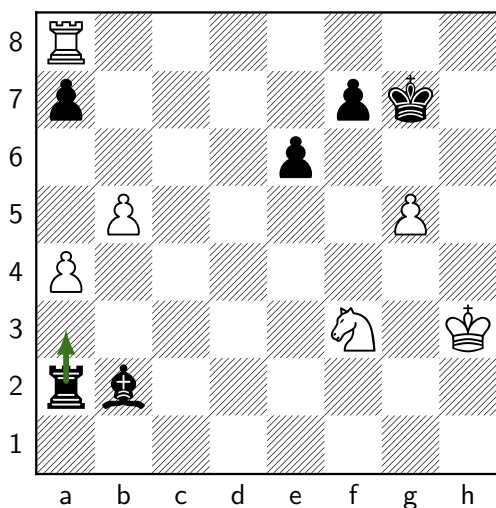
Figure 37: **Move 43, Black: Kh7**. (a) Our position; Melanie plays Kh7 (g6h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1815` (ply 71) [structural distance $d=28$, QAP obj=69]: the master played Kg7; under μ our Kg3 plays the role of that game’s Kh1; cross-type roles: our be5 as their Ph7, our Pe6 as their nd6; transferred move g6h7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1763` (ply 72) [structural distance $d=29$, QAP obj=51]: the master played Kh2; under μ our Kg3 plays the role of that game’s Kh1; cross-type roles: our be5 as their ne3, our Bc8 as their Pg3; transferred move g6h7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1104` (ply 87) [structural distance $d=28$, QAP obj=31]: the master played Kf7; under μ our Kg3 plays the role of that game’s Kf1; cross-type roles: our Bc8 as their Pa7, our Pd5 as their Bb7; transferred move g6h7 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Kh7 (summed QAP objective 349, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



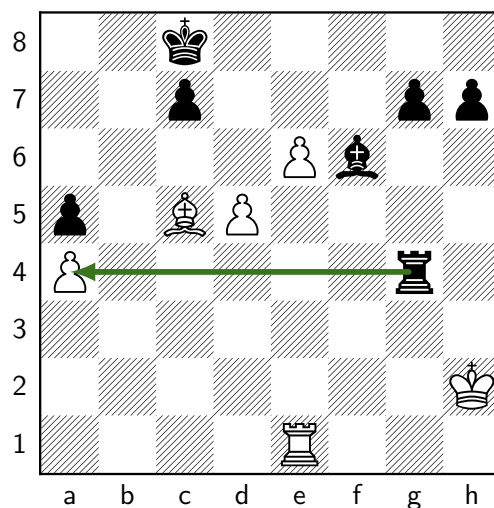
(a)



(b)

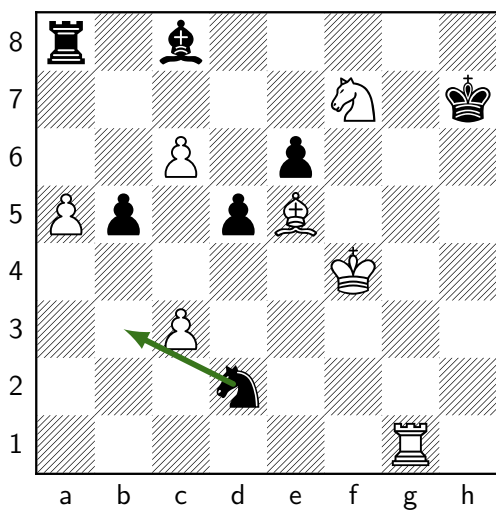


(c)

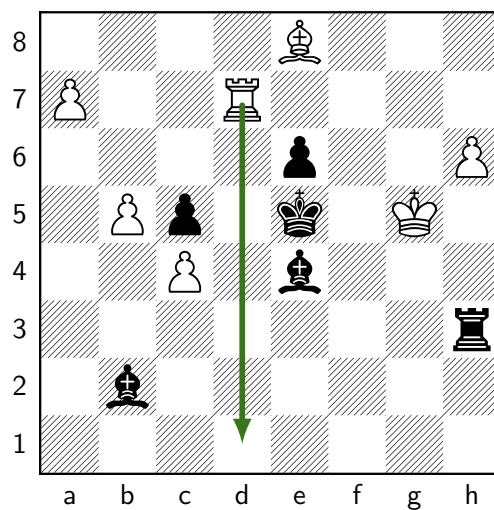


(d)

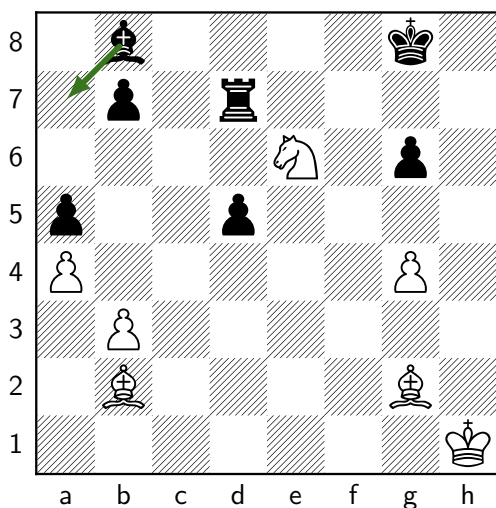
Figure 38: **Move 44, Black: Nd2+.** (a) Our position; Melanie plays Nd2+ (b3d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#934` (ply 86) [structural distance $d=25$, QAP obj=12]: the master played Ng7; under μ our Nb6 plays the role of that game’s Nh5; cross-type roles: our Pd5 as their Rc1, our Ra8 as their Bd4; transferred move b3d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#290` (ply 85) [structural distance $d=27$, QAP obj=12]: the master played Ra3; under μ our Nb6 plays the role of that game’s Ra7 (a **cross-type** correspondence); cross-type roles: our nf7 as their pg5, our pc6 as their nf3; transferred move b3d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#809` (ply 91) [structural distance $d=30$, QAP obj=10]: the master played Rxa4; under μ our Nb6 plays the role of that game’s Rg5 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Pg7, our nf7 as their Ph7; transferred move b3d2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nd2+ (summed QAP objective 45, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



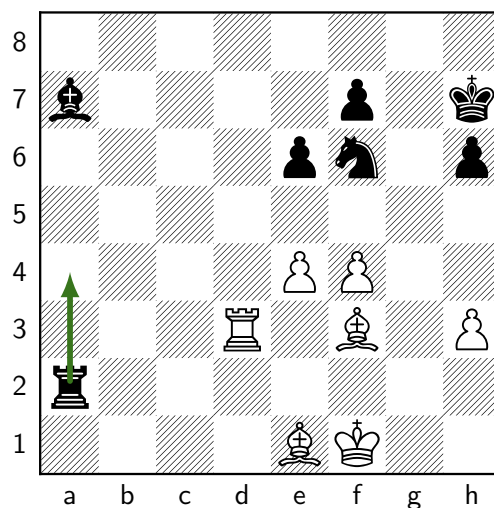
(a)



(b)



(c)



(d)

Figure 39: **Move 45, Black: Nb3.** (a) Our position; Melanie plays Nb3 (d2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3142` (ply 104) [structural distance $d=35$, QAP obj=21]: the master played Rd1; under μ our Nd7 plays the role of that game's Rd7 (a **cross-type** correspondence); cross-type roles: our nf7 as their Pa7, our rg1 as their Be8; transferred move d2b3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2879` (ply 89) [structural distance $d=34$, QAP obj=21]: the master played Ba7; under μ our Nd7 plays the role of that game's Bb1 (a **cross-type** correspondence); cross-type roles: our nf7 as their Pb7, our be5 as their Rd7; transferred move d2b3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3886` (ply 75) [structural distance $d=31$, QAP obj=14]: the master played Ra4; under μ our Nd7 plays the role of that game's Ra7 (a **cross-type** correspondence); cross-type roles: our Pd5 as their Ba7, our Bc8 as their Pe6; transferred move d2b3 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Nb3 (summed QAP objective 103, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.